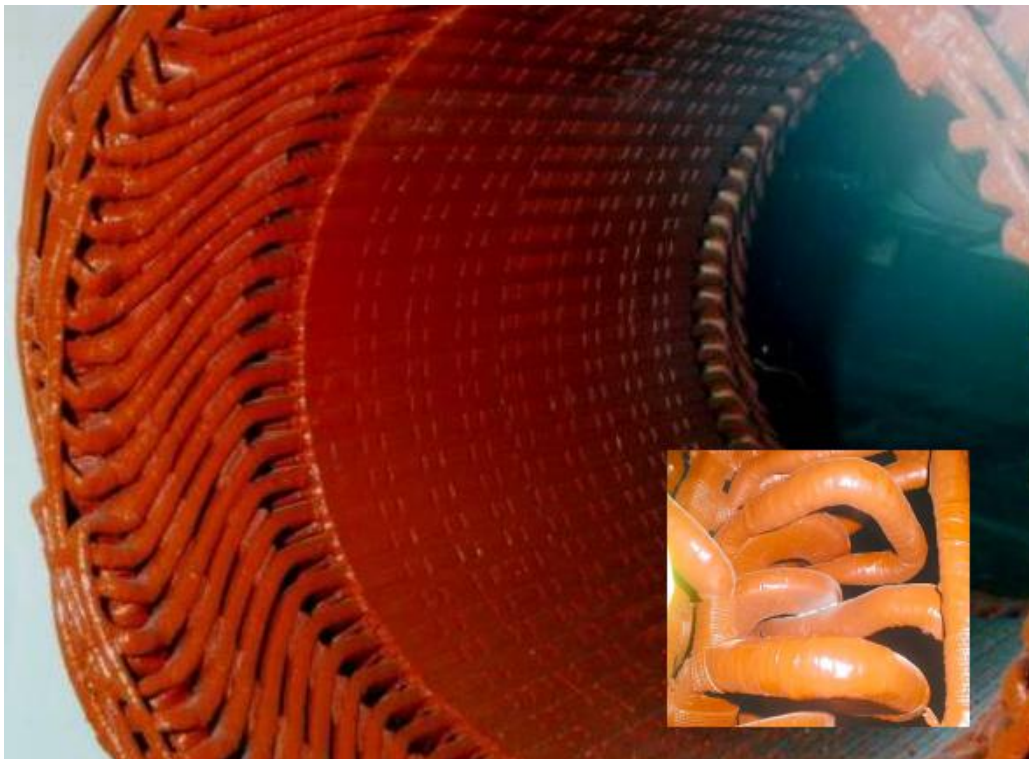


Interpretation of Partial Discharge Results

Electrical Rotating Machine Stator Windings

Document ID: PD Report Appendix, Version 4





Interpretation of PD Results – Electrical Rotating Machine Stator Windings

May 2013 - Version 4

Information in this document is subject to change without notice. This document is provided to purchasers of Iris Power products and service for use of such products and service. No other use, nor any reproduction, distribution or the making of any derivatives of this document is authorized, without the express prior written permission of Iris Power.

Iris Power endeavors to ensure the accuracy and quality of its published materials; however, no warranty, expressed or implied, is provided. Iris Power disclaims any responsibility or liability for any direct or indirect damages resulting from the use of the information in this manual or products described in it. Mention of any product or brand does not constitute an endorsement by Iris Power of that product or brand.

This document was originally composed in English. The fidelity of any subsequent translations into other languages cannot be guaranteed. In case of conflict between the English version and another language version, the English version takes precedence.

©2013 Iris Power L.P., an ISO 9001 system certified company. All rights reserved. Information is subject to change without notice.

All trademarks are properties of their respective companies, as noted herein.

Windows of any versions are products of Microsoft® Corporation.

PD LitePro™, PDView™, TGA™, PDA-IV™, Guard™, BusGuard™, HydroGuard™, TurboGuard™, GuardII™, Trac™, BusTrac™, HydroTrac™, PDTrac™, BusTracII™, HydroTrac™, PDTrac™, and Portables™ are products of Iris Power L.P.



TABLE OF CONTENTS

1 INTRODUCTION	3	4.3.2 Phase-to-Phase PD	17
2 DETECTION OF PD	4	4.3.2.1 Contamination	17
2.1 Capacitive Couplers	4	4.3.2.2 Inadequate Spacing	18
2.2 Stator Slot Couplers (SSC)	4	4.3.2.3 Gap-type Discharges	18
3 INTERPRETATION	5	4.4 Other Arcing and Sparking	19
3.1 Trend analysis	5	4.4.1 Broken Strands Arcing	19
3.1.1 Long-term trend	5	4.4.2 Vibration Sparking	19
3.1.2 Short-term movement	7	4.5 Extraneous Signals.....	20
3.2 Database Comparison	7	4.5.1 Excitation pulses.....	20
3.3 Phase Comparison	8	4.5.2 Cross-coupling.....	20
3.4 Operating Condition Trend	8	4.5.3 Broadband Clusters.....	21
3.4.1 Load Effect	9	4.5.4 High level disturbance	21
3.4.2 Temperature Effect	9	4.6 Summary - Failure Processes.....	22
3.4.3 Hydrogen Pressure Effect	10	5 APPENDIX.....	23
3.4.4 Humidity Effect	10	5.1 Noise and Disturbances	23
3.5 Disturbances and Noise	10	5.1.1 Noise Suppression	23
4 FAILURE MECHANISMS	11	5.1.2 Disturbance Separation	23
4.1 Multiple Failure Mechanisms	11	5.1.2.1 Impedance Mismatch.....	24
4.2 Discharges in the Slots	11	5.1.2.2 Time Domain Separation	24
4.2.1 Internal Discharge	11	5.1.2.3 Pulse Shape.....	25
4.2.1.1 Internal Voids and Delamination	12	5.2 Data Types	26
4.2.1.2 Manufacturing Defect	12	5.2.1 PD Summary Variables	26
4.2.1.3 Delamination between Conductor and Insulation	13	5.2.2 Phase Angle Resolved Plots	27
4.2.2 Slot Partial Discharge	14	5.3 Zone of Coverage	30
4.2.2.1 Bar and Coil Movement	14	5.3.1 TGA - SSC Test Method	30
4.2.2.2 Electrical Slot Discharge	15	5.3.2 TGA - BUS Test Method	30
4.3 Discharges in the Endwinding	15	5.3.3 TGA - Single-ended Test Method.	31
4.3.1 Surface Discharge at Voltage Stress Control Coatings Interface	16	5.3.4 PDA Test Method	31
		5.4 Prime (Alternate) Numbers	31
		5.4.1 PD in the Slots.....	32
		5.4.2 Prime (Alternate) Numbers.....	33
		6 REFERENCES	34
		7 GLOSSARY.....	35



1 INTRODUCTION

For over 60 years, it has been known that Partial Discharge (PD) is a symptom of several problems caused by thermal, mechanical (vibration and shock), electrical (voltage), environmental, and chemical stresses [1, 2]. Rarely would any one factor occur alone; stresses combine leading to many different deterioration processes.

Partial discharges (PD) are small electrical sparks that occur when gas pockets exist within high voltage insulation [1]. Gas pockets can occur because of the manufacturing process, thermal deterioration, stator winding movement during operation and other possibilities. As the insulation degrades, the number and magnitude of the PD will increase. These happen because the breakdown strength of air (3 kV/mm peak) or hydrogen is much lower than that of solid insulation (about 300 kV/mm peak). The breakdown of gas inside the pockets creates small voltage pulses that can be detected and measured, and therefore the PD can be monitored. Although the magnitudes of the PD pulses cannot be directly related to the remaining life of the winding insulation, the rapid increase in the rate of PD activity or high PD levels compared to other similar windings are indicators that visual inspections and/or other tests are needed to confirm the insulation condition. Hence a PD test assists companies to plan the maintenance requirements of machine stator windings. For information regarding inspections and other suitable tests, refer to *Electrical Insulation for Rotating Machines* [2].

Deterioration of the organic resins in the insulation materials due to PD is an extremely slow process since most stator winding insulation systems for machines rated greater than 2300 V contain a discharge-resistant material called mica; however, the prediction of time to failure is impossible. Eventual insulation failure will occur during a switching surge, poor synchronization, load rejection or some other event causing insulation breakdown at the weakest point. On-line PD tests should be conducted at regular intervals and, where appropriate or indicated, be supplemented with visual inspections and off-line testing. Consistent and regular testing will provide the means to assist plant personnel in determining the overall winding insulation condition. The plant's history of a machine could be beneficial in deciding the proper course of action.

Because of this relatively slow ageing process, periodic monitoring of the PD activity makes sense. The rule of thumb regarding the length of time between detection of significant PD and insulation failure due to damage by PD is:

- 10 years for machines over 18 kV
- 5 years for 13.8 kV machines
- 2 to 3 years for 6 kV machines
- Several months for 3.3 kV and 4 kV machines

The terms “coils” and “bars” are used interchangeably in the text and either of them means the stator winding.

2 DETECTION OF PD

PD detection involves measuring four characteristics of the PD patterns:

- PD magnitude $\Rightarrow$ relates to the size or volume of the defects
- PD pulse count rate $\Rightarrow$ relates to the number of defects
- PD polarity $\Rightarrow$ relates to the location of defects within the slot section of winding
- PD position relative to the phase-to-ground voltage $\Rightarrow$ relates to the location of defects either in the slot or at the endwinding

Permanently mounted PD sensors attenuate the AC power signal, but pass the high frequency PD pulses. During normal operation, an Iris instrument connected to the sensors separates electrical noise and disturbances [12], thus allowing proper classification of the PD. The type of sensor installation and test instrument depend on the machine or equipment being monitored. Iris uses two types of sensors:

- Capacitive couplers, or Epoxy Mica Capacitors (EMC) for motors, hydro generators, and small turbine generators.
- Stator slot couplers (SSC) - for large turbine generators (Usually larger than 100 MW).

Each sensor installation and test instrument classify pulses differently based on the **Zone of Coverage** – see Section 5.3 for more information.

2.1 Capacitive Couplers

The capacitive couplers used are $80 \text{ pF} \pm 4 \text{ pF}$ (Figure 1) and they meet all the requirements of IEC 60034-27-2 [12]. The 50/60 Hz signal is attenuated and the high frequency PD signal can pass more easily. Elegantly simple in appearance and construction, these coupling capacitors are rugged and reliable, presenting little risk to the machine on which they are installed. They are made to close electrical tolerances and of necessity, they are partial discharge free well beyond voltage levels normally encountered in machine operation. These capacitors are permanently attached to the phase terminals, the phase bus-bar or the circuit ring. By connecting the couplers to these high voltage points, they are close to the areas in the winding that have the highest partial discharge activity. Remember, partial discharge is dependent upon the applied voltage, so only the line-end coils are likely to have discharge activity.



Figure 1: EMC couplers

2.2 Stator Slot Couplers (SSC)

On many large (over 100 MW) high-speed turbine generators, there are PD-like pulses (disturbances) that occur because of core iron arcing and sparking due to poor connections at the machine terminals. It is impossible to discriminate between these pulses and true (insulation) PD using external capacitive couplers, so it is recommended that an antenna type sensor be used. This antenna sensor, called stator slot coupler (SSC), is comprised of an electrode structure printed on an epoxy-fiber glass laminate (Figure 2).

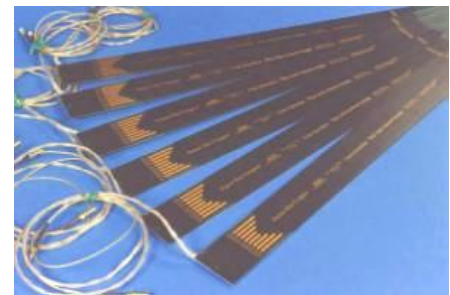


Figure 2: Stator Slot Couplers (SSC)

The SSC is placed in the slot, either embedded between the top and bottom bars or directly underneath the wedge. Like the capacitive couplers, the SSC attenuates the low-frequency noise pulses but detects the high-frequency PD pulses. Since the SSC makes no electrical connection to the winding, it is only sensitive to PD occurring within the slot containing the SSC. Though this limits the amount of coverage for the sensor, it does guarantee that only PD from the slot and not high-frequency disturbances from the core, brushgear or other local sources will be detected. When the SSCs are installed within a hydrogen environment, a pressure penetration flange is required. In addition, an external voltage reference is needed to locate the PD pulses relative to the AC phase-to-ground voltage since the SSC is not connected to the winding.

3 INTERPRETATION

3.1 Trend analysis

The first step of analysis is comparing the results of the current test with any previous test results. Trending of the Q_m and NQN values can give an indication of the *progression* of the ageing mechanisms. A doubling of PD activity (Q_m values twice that of the previous test) every six months is a strong indication of a rapidly developing failure mechanism [11]. (See Section 5.2.1 for the explanation of the PD Summary Numbers: Q_m and NQN)

For our purposes, Q_m values are derived from the test result and are valuable for comparisons amongst machines or trending over time. There are three significant trends, the *short-term*, *long-term* and *operating condition trend*. See Section 3.4 for more information about the operating condition trend.

If the unit operating parameters – voltage, load, winding temperature and gas pressure – are similar to those of the previous test, then a direct comparison can be made between the two test results. For cases where the operating parameters are not the same, refer to Section 3.4. In order to compare tests properly, it is best to maintain operating conditions within certain ranges, such as:

Voltage	$\pm 2.5\%$	PD increases with voltage, so variances in voltage can lead to fluctuations in PD levels.
Load	$\pm 10\%$	If a machine has PD that is mechanically dependent, such as loose coils or connections, then it is likely the PD levels will change with load.
Real power	$\pm 10\%$	
Reactive power	$\pm 10\%$	
Stator temperature	$\pm 10^\circ\text{C}$	Operating temperature affects the volume of internal voids, which changes PD magnitudes.
Hydrogen pressure	$\pm 5 \text{ psi-g}$ $\pm 35 \text{ kPa}$ $\pm 0.35 \text{ bar}$	Gas pressure impacts the electrical breakdown stress and therefore levels of PD.
Humidity	Unknown	Humidity can impact surface PD, especially PD at the endwinding or on ring busses

3.1.1 Long-term trend

Purpose: To determine whether there is a notable long-term trend over many years that is changing.

Requirements: Linear regression comparison to the “best” measurements from all relevant test dates within the database for a specific asset. Note that tests older than the winding itself should be excluded and data collected within the first 12-18 months of operation may be excluded if they highly influence the results. The latter is due to the common occurrence that PD behaviour is often erratic and may show a gradual reduction for new windings as the winding settles-in and the bonding resin continues to cure.

Long-term trends may not be possible if the current test results are outside of the norm, that is, judged to be unreasonably high or low relative to the trend. The trend will be described as “Unknown.”

When a trend line is established for PD tests taken over a longer period, it is likely that most trends show an up and down variation between successive tests. However, as an insulation system ages, there will be an easily discernible overall upward movement of PD with time if the PD is measured at approximately the same operating conditions. Ageing is normally a very slow process and sudden changes are not expected in the PD test results. Though the condition of the stator winding insulation can be assessed, time to failure cannot be predicted. The actual failure is normally the result of an unusual source of insulation stress such as lightning, out-of-phase synchronization or severe overheating.

Maintenance can often be done on a machine to lower the PD activity. Examples of maintenance that have been known to successfully reduce PD are re-wedging, cleaning, dip and bake, and repairs to the voltage stress coatings. If the source of the PD is within the bulk of the insulation, then repairs may not be effective.

Caution: Erratic PD can cause wide swings in trends that may be misleading. Do not interpret these in isolation. Variations of some percent, say +/- 25%, are normal [11].

Long-term trend analysis:

<i>Baseline</i>	No previous results exist	
<i>Stable $\pm 25\%$</i>	Slope of the line and change within an expected range	No significant change in activity
<i>Slow upward</i>	Minor, but notable, increasing slope over several months	Not significant, but should be closely monitored if sustained
<i>Upward trend</i>	Minor, but notable, increasing slope over limited months	
<i>Downward trend</i>	Minor, but notable, decreasing slope over several months	
<i>Significant increase</i>	Increasing slope over limited months	Significant changes are unusual, and should be closely monitored during the next testing cycle. Retest is highly recommended.
<i>Significant decrease</i>	Decreasing slope over limited months	
<i>Rapid deterioration</i>	Doubling of PD in 6 months or rapid increase over a short interval	Should be retested and perhaps supplemented with offline tests and visual inspections.
<i>Unknown</i>	The Q_m values are too volatile to be evaluated. This normally occurs when the current test results are outside of what would be considered reasonable for a consistent trend.	

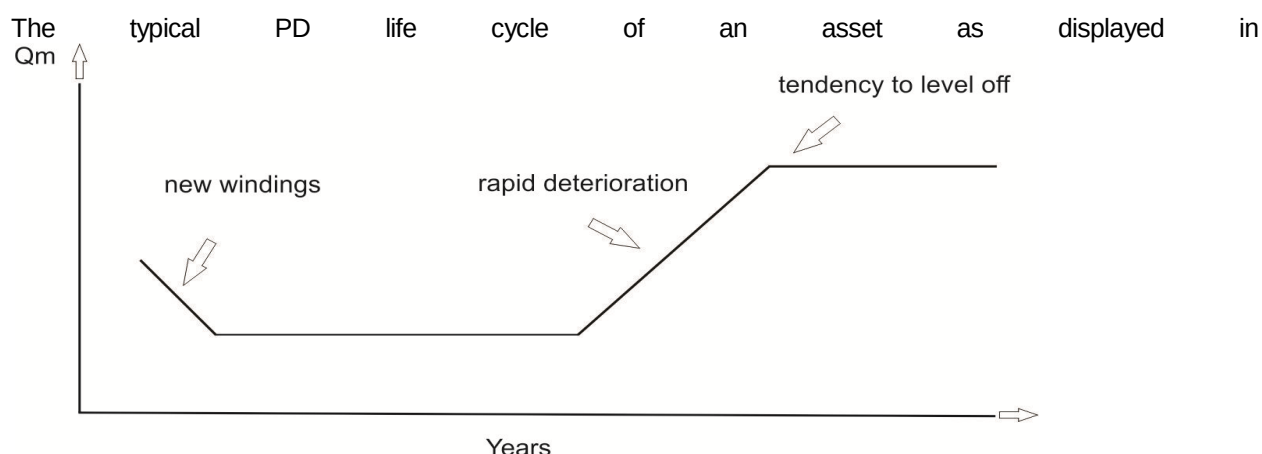


Figure 3 shows that on a new machine the PD will often start relatively high and then decrease as the winding settles over the first 12 months to “baseline” levels [8]. Following that, there should be many years of relatively stable levels, with perhaps minor fluctuations due to the influence of variable ambient and operating conditions. Once a problem develops, the PD levels will increase quite rapidly, and then may stabilize at a high level due to several physical processes [2].

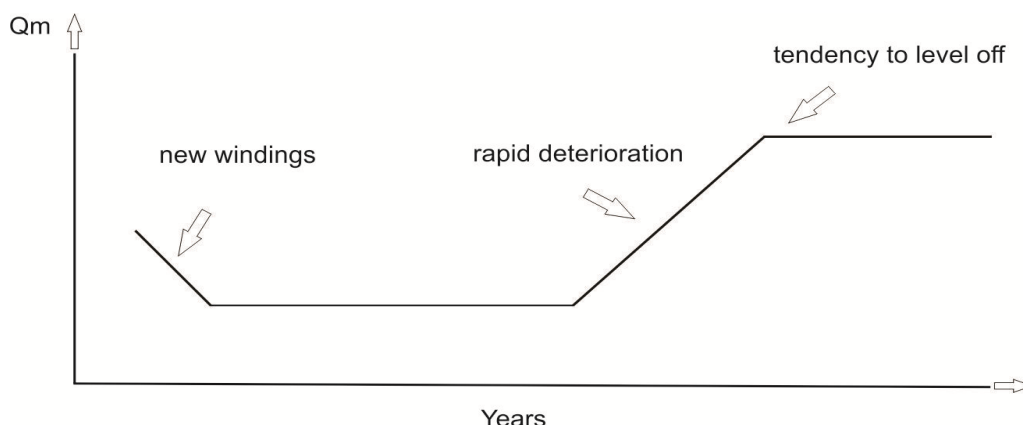


Figure 3: PD Life Cycle

Should it be possible to repair the damage, then the levels should decrease back to those observed prior to the onset of the problem, and the process repeats – with many years of relatively stable activity until the PD levels once again start increasing. As previously stated, there is no way to predict when failure will occur, as failure normally happens when a compromised insulation system is exposed to an unusual, random stress. Based on the insulating materials, and nature of the deterioration, an inexplicable decrease or shift in the PD patterns may also be an indication of a rapidly developing problem.

Note PD may not continue to increase until failure occurs. Any significant change in activity, up or down should be investigated [1].

3.1.2 Short-term movement

Purpose: To discern whether there has been a sudden and significant change from the previous test

Requirements: Comparison to the most recent tests which are at least 1-month old. It is preferable that the previous results are not older than 6 months from the current test; however, evaluations are done regardless of this requirement.

Changes in PD levels are usually slow to develop. If the operating conditions are consistent, significant variation up or down over a short time period is cause for attention and indicates a need to retest as soon as possible to determine whether the change is permanent or intermittent. Maintenance decisions should not be made in isolation based on a single change in pattern – PD is stochastic in nature and can highly be influenced by variances in operating and ambient conditions [Section 3.4].

Short-term trend analysis:

<i>Baseline</i>	No previous results exist	
<i>Stable $\pm 25\%$</i>	Within plus or minus 25% of previous results	No significant change in activity
<i>Upward trend</i>	Minor, but notable, increase	Not significant, but should be closely monitored if sustained
<i>Downward trend</i>	Minor, but notable, decrease	
<i>Significant increase</i>	Increase from previous results	Significant changes over the short-term are unusual, and should be closely monitored during the next testing cycle. A retest is highly recommended.
<i>Significant decrease</i>	Decrease from previous results	

3.2 Database Comparison

Purpose: To determine whether the results are in-line with similar machines

Requirements: Comparing the highest Q_m value relative to the results published regularly by Iris Power L.P.

As mentioned in Section 3.1, data collected from the same machine operating under similar parameters using the same test setup are directly comparable and the most useful. Comparisons to similar stator windings indicate the *relative severity* of failure mechanisms. It is also possible to roughly compare the results of one machine to those of similar machines.

Levels of comparisons with similar stator windings:

Negligible: Q_m below 25%
 Low: Q_m between 25% and 50%
 Typical: Q_m between 50% and 75%
 Moderate: Q_m between 75% and 90%
 High: Q_m between 90% and 95%
 Very High: Q_m Over 95%

Due to the influence of the test frequency on the results, the test setup (sensors and test instrument) must be the same for all comparisons. Based on statistical analysis, the type of machine does not appear to have a great impact to PD levels. In addition, although insulation types have different types of failure mechanisms, there does not appear to be much difference in the overall levels of PD activity [11] in these gross comparisons. Load and temperature can also greatly influence test results, but these are dependent upon the type and condition of the insulation system, and thus, cannot be used in these gross comparisons. The remaining operating variables – voltage and gas coolant pressure – must also be similar to compare results [11].

As described in Section 5.1.2.2, the coupler location and pulse classification are based on the position of the coupler relative to the winding. Interpretation should always consider the **Zone of coverage** concept; so that visual inspections will start at machine terminals when appropriate (Section 5.3).

A further caution that can have an impact on comparison of PD data from similar machines and the Iris statistical database needs to be highlighted [11]. Like all current pulses, the high characteristic frequency partial discharges are greatly affected by impedance mismatches in the path they traverse. If accessibility issues cause the PD sensors to be located away from the machine terminals or winding, and there are locations of impedance mismatches between the sensor and machine or winding, then the pulse magnitudes will be affected. However, this will have no impact at all on the Trend evaluation, which is the primary purpose of the partial discharge test, but it could influence the ability to compare results to the Iris statistical database.

Because of the great variety of design features, production conditions and insulation systems, it is not possible to directly compare the absolute values for different types of machines or to establish absolute acceptance limits [10]. The statistical comparison suggested here is based on gross comparisons only.

3.3 Phase Comparison

Purpose: To determine if the PD levels are comparable among the phases

Requirements: Evaluate the Q_m values from each phase for relative magnitude

When the PD levels and patterns are similar among the phases, then *uniform* deterioration is suspected; otherwise, an *isolated problem* is more likely. Typically, coil looseness, thermal deterioration, contamination, and problems with the voltage stress coatings are widespread, or *uniform*, whereas, poor manufacturing consistency and inadequate spacing are often *isolated problems*.

3.4 Operating Condition Trend

Purpose: To determine if the PD Q_m changes with differences in operating conditions, ambient conditions or both



Requirements: Similar to the long-term trend, a linear regression analysis based on results extracted from the entire database that have similar operating condition, ambient conditions and test sensitivities.

Environmental conditions such as humidity may have a very noticeable impact, especially if the surface contamination to some extent becomes conductive when damp, so the humidity should be recorded from one test to the next. The recording will assist in determining if swings in PD may be attributable to changes in humidity if all other operating parameters are similar. It is appreciated that at times it may be very difficult to duplicate test conditions, particularly temperatures; but the emphasis should always be on trying to achieve as consistent conditions as possible. This means taking tests when units are hot and the temperature of insulation has stabilized. Due to thermal inertia of organic stator insulation, this usually happens after some time elapses from the point when RTD temperature readings became stable. When unloading a machine to do a reduced or no-load test, it is inevitable the temperature will be in a state of transition through the test. In these cases, it is important to quickly conduct the tests to minimize the effect of temperature differences.

PD is very dependent on voltage, so small swings in the voltage can lead to changes in the PD results. When the predominant problem is due to voids within the groundwall insulation due to [Manufacturing Defects](#), [Internal Delamination](#) and [Thermal Cycling](#), the tests at lower operating temperatures (cold test) tend to have higher PD levels than the tests at higher temperatures (hot test). This phenomenon may occur because the copper and insulation expand at the higher temperatures, therefore decreasing the void volume and the resultant PD [see Table 3-1].

If the predominant problem is due to mechanical stresses, such as [Coil/Bar movement](#), PD activity at lower loads (or no load) will be significantly lower than tests at higher loads (full load) perhaps because there are smaller mechanical forces present at the lower load. If the predominant problem is [Phase-to-Phase PD](#) or due to [Contamination](#), surface humidity can greatly affect the results. Since the occurrence of PD can vary due to small changes in ambient or operating condition, caution must be taken that successive tests be carried out under similar conditions.

Table 3-1. Operating Condition Effects

Temperature effect	Direct (Positive)	Increase in PD at higher temperatures
	Inverse (Negative)	Decrease in PD at higher temperatures
Load effect	Direct (Positive)	Increase in PD at higher loads
	Inverse (Negative)	Decrease in PD at higher loads
Hydrogen pressure effect	Direct (Positive)	Increase in PD at higher pressures
	Inverse (Negative)	Decrease in PD at higher pressures
Humidity effect	Direct (Positive)	Increase in PD at higher humidity
	Inverse (Negative)	Decrease in PD at higher humidity

3.4.1 Load Effect

For [Coil/Bar Movement](#) (loose windings), PD can be very load-dependent with positive PD activity increasing with the load, *direct effect*. This may be due to the increase in magnetic forces ($F_m \propto I^2$) causing an increase in coil movement and thus surface (positive) PD activity. The effect is most significant with loose thermoset (hard) windings such as epoxy-mica insulated windings. Thermoplastic (soft) windings expand when heated and tend to secure themselves within the core. If PD load dependency is present, it is always a concern that loose coils can lead to premature winding failure. These coil or bar movements, if present, can rapidly damage the resistive surface coating on the coil sides, and if left unchecked, they can wear away the insulation itself. In extreme cases, this failure mechanism can destroy an otherwise good winding in only a few years. Be aware that the initial stage of looseness is called “contact sparking,” and can often happen on coils throughout the winding, not just the high voltage ones. The load effect may also occur with the vibration sparking (or spark erosion) failure mechanism [13].



3.4.2 Temperature Effect

Operating temperature can greatly influence the results from a partial discharge test. Various materials respond differently to changes in temperature. These changes can be subtle for small variances of ± 5 °C, or drastic, doubling and tripling, for greater discrepancies of ± 20 °C. The sizes of voids within the stator winding insulation are usually inversely proportional to the operating temperature. As the temperature increases, the copper and groundwall bonding material expand closing voids and thus decreasing the PD magnitude of internal sources [1]. This *inverse effect* is more significant on asphalt and polyester windings, but can also be observed on epoxy windings. A large inverse temperature effect suggests more severe internal delamination. Note that stator slot RTD temperatures do not quickly reflect changes in the temperature of the copper conductor or the internal insulation. It may be possible that positive PD actually increases with temperature (*direct effect*). This phenomenon is frequently an indication of deterioration of the voltage stress coatings [1]. This problem is generally a very slow failure mechanism, but often leads to high ozone production in air-cooled machines.

3.4.3 Hydrogen Pressure Effect

Since the occurrence of partial discharge is greatly affected by the shape and gas pressure within the void, variances in operating hydrogen pressure can lead to high fluctuations of the overall PD activity. These changes can be subtle, for small variances of ± 2 psi, or drastic, doubling and tripling, for greater differences of ± 5 psi. Performing the PD test with the machine at the same load, but at different operating pressures, that is at least ± 10 psi apart, may be able to help identify which failure mechanisms are occurring. If the PD decreases with small increases in the hydrogen pressure (*inverse effect*), then the sources of PD are most likely within the hydrogen environment. If there is no noticeable change in the PD pattern at different hydrogen pressures (*none*) or a decrease in PD at lower pressure (*direct effect*), then it is likely the PD is from sources external to the hydrogen environment, such as at the machine terminals. Surface discharges are more prominent at low gas pressure [1].

3.4.4 Humidity Effect

Perhaps the most influential, yet unpredictable, impact of ambient conditions to PD is the effect of variances in ambient humidity on air-cooled machines. Humidity can cause the electrical breakdown stress of the air to decrease, and therefore cause an increase in surface PD. Humidity can affect the electrical tracking of surface PD, and therefore cause a decrease in PD. Thus, in some circumstances, the PD may go up, *direct effect*, while in others it would go down with increases in relative humidity, *inverse effect*. In some circumstances, the humidity effect has been known to cause the PD to fluctuate by as much as 300%. Since each scenario is different, it is extremely important that the ambient humidity be recorded at test time, so trends can be properly evaluated and decisions are not made about upward trends that occur because of humidity effects and not changes in winding condition. Present experience indicates humidity only affects surface activity [1].

3.5 Disturbances and Noise

Most companies prefer a PD test that can be performed by their staff during normal machine operation. Electrical noise from power tool operation, corona from the switchgear and radio frequency sources etc. are easily confused with PD from the machine windings. This confusion can lead to healthy windings being misdiagnosed as deteriorated and thus lowering confidence in the test results. A dependable on-line PD test should significantly reduce the influence of noise and disturbances, thus leading to a more reliable indication of the insulation condition [3].

IEC 60034-27-2 defines that “*Disturbances* are electrical pulses of relatively short duration that may have many of the characteristics of stator winding PD pulses – but in fact are not stator winding PD” [12]. Some of these disturbances are synchronized to the AC cycle and some are not. Sometimes synchronized disturbance pulses can be suppressed based on their position with respect to the AC phase angle. See Section 5.1.2 for more information about Disturbance separation.



“Noise” is defined to be “non-stator winding signals that clearly are not pulses” [12]. Electrical signals resulting from power tool operation, corona from the switchgear and radio frequency sources are all examples of such noise. Noise may also be due to electronic devices within the PD detection system itself. See Section 5.1.1 for more information about Noise suppression.

4 FAILURE MECHANISMS

4.1 Multiple Failure Mechanisms

PD plots may be complicated if two or more failure mechanisms or processes are occurring in the winding at the same time. For example, contamination and a loose stator winding could be occurring simultaneously. In these situations and in the presence of high PD magnitudes, subsequent PD testing, off-line PD tests, other electrical tests and visual inspection may be needed to conclusively separate the different failure processes.

Below are descriptions of the most common stator winding insulation problems viewed in isolation. Two or more such deterioration processes may superimpose PD pulses in a complex pattern.

4.2 Discharges in the Slots

Inspection of the PD data reveals the phase angle relationship between the PD pulse and the phase-to-ground reference voltage. PD that originates within the slot portion of the coil or bar is centred at 45° for the negative pulses and 225° for the positive pulses (Figure 4). These are the *phase-to-ground PD positions* for pulses dependent upon the phase-to-ground voltage of the specific phase being measured. Phase-to-ground voltage stresses occur all along the coils as they pass along the length of the slots in the stator core. Hence, this type of pattern is indicative of PD occurring in the slots and is referred to as “PD in slots” or phase-to-ground PD.

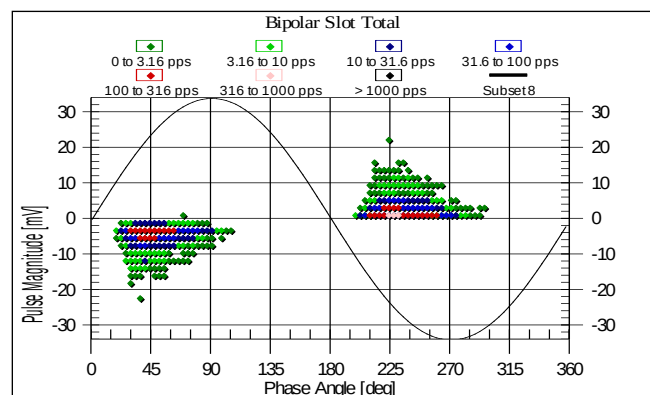


Figure 4: PD in Slots

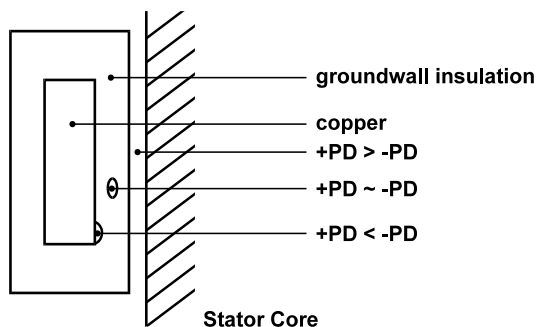


Figure 5: PD Pulse Polarity

Modelling the actual characteristics of a pulse is difficult since the void dimensions, void gas and pressure, inductance, capacitance, geometry and many other issues can affect the magnitude and frequency of a pulse [1]. However, some basic pulse characteristics can be predicted based on the void location as shown in Figure 5.

Positive predominance is an indication of surface PD due to coil/bar movement, electrical slot discharge or arcing at the voltage stress coating. *Negative predominance* indicates internal PD from delamination, damage due to thermal cycling or excessive starts and surges, or inadequate bonding during manufacturing.

4.2.1 Internal Discharge

Windings that have *not been properly impregnated* or that have been operating for several years at *high temperatures* tend to develop voids within the groundwall insulation. Partial discharge will occur across such a void when both an over voltage condition and a free electron are present. Once the partial discharge occurs, the voltage across the gap will stabilize at the level required to sustain the discharge activity. During this process, the charges within the void will redistribute according to the applied charge.

As the AC cycle reverses in polarity, these charges will cause another over voltage condition in the opposite polarity to be established and another partial discharge may occur. If both sides of the void have similar insulation materials then the charge distribution will be equal during the positive and negative cycles [1]. In theory, as shown in Figure 6, there will be two observable PD pulses in each AC cycle of *equal magnitudes* and opposite polarity per void within the bulk of the insulation. These pulses clump at the typical positions for phase-to-ground voltage dependent pulses, that is, *negative pulses at 45°* and *positive pulses at 225°* with reference to the 50/60 Hz phase-to-ground voltage. Please note that pulses can occur at opposite polarity of these positions in some circumstances, see Section 5.4 for more information.

Deterioration from internal voids is usually a slowly developing long-term ageing mechanism.

4.2.1.1 Internal Voids and Delamination

Internal Delamination can occur as a result of long periods of overloads, defective cooling, unbalanced phase voltages and poor design. As with most insulation systems, the damage is cumulative, non-reversible and results in decreased ability of the resin binder (epoxy, polyester polyester or asphalt) to mechanically bond the layers of insulation together. This loss of mechanical bonding allows the formation of voids within the layers of tape that make up the insulation thickness. As the tape layers separate, PD is created in the voids and the conductors can become free enough to vibrate. The PD, along with possible mechanical abrasion, may lead to strand and turn shorts.

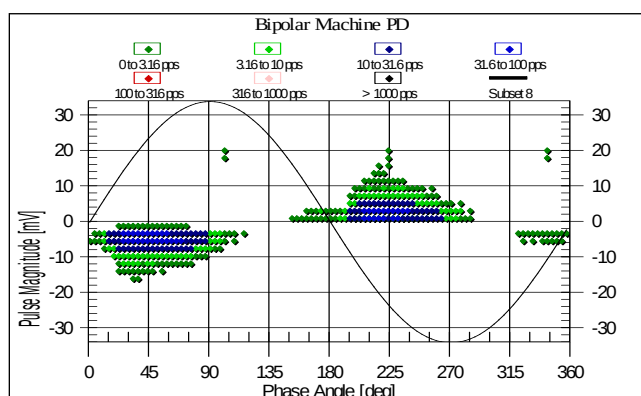


Figure 6: Internal Delamination or Manufacturing Defect

Initially, these voids may be near the conductors and the PD exhibits a *negative* predominance, but over time these voids are distributed throughout the insulation and may discharge regardless of the “direction” of electrical stress across them. Because of this, such voids may not exhibit any PD polarity predominance and the PD clusters near 45° and 225°. Therefore, a fairly equal distribution of positive and negative discharges at these phase positions is indicative of general internal delamination or ageing (Figure 6).

Due to the difference between the thermal coefficients of expansion of copper and insulation, the voids originating in the slot portion usually reduce in volume as the winding temperature increases, producing an *inverse temperature effect*. The extent of this change in PD with temperature is dependent on the degree of delamination that may be present. There is usually minimal change in PD with load.

Failure mechanism	Polarity	Load Effect	Temperature Effect	Phase Location
Internal Delamination	No or Negative predominance	None	Inverse	45° and 225°

4.2.1.2 Manufacturing Defect

Sometimes during the impregnation process, small voids are inadvertently left within the groundwall. These can occur because of inadequate vacuum or pressure, incorrect resin viscosity, improper temperatures, tape wrinkling or foreign objects imbedded in the tapes. In the presence of voltage stress, partial discharges can occur across these voids and attack the organic resin. The attack may lead to strand or turn shorts in multi-turn coils if the voids are near the copper conductors, and eventually failure of the coils can occur.

If the voids are in the centre of the groundwall insulation thickness, they are more benign and low magnitude, so failure can take many years to happen. An additional problem from manufacturing defect is that the internal voids create a thermal barrier inhibiting the transfer of heat from the copper to the core resulting in higher thermal stresses.

Like internal delamination, these voids may be near the conductors or are often scattered throughout the insulation without dominance of voids either close to the copper or at the surface. There may be a slight negative or perhaps no noticeable polarity predominance of the PD with activity centred near 45° and 225°. As stated before when describing internal delamination, internal voids usually lead to a decrease in PD with an increase in temperature, that is, an *inverse temperature effect*, but no changes with load (Figure 6).

Failure mechanism	Polarity	Load Effect	Temperature Effect	Phase Location
Manufacturing Defect	No or Negative predominance	None	Inverse	45° and 225°

4.2.1.3 Delamination between Conductor and Insulation

A machine that is frequently *thermal-cycled* or *severely overheated* develops voids near the copper conductors. The negative impact of frequent changes to the machine load is the cyclical shear stresses placed on the insulation due to different linear coefficients of thermal expansion in the stator winding materials. As the copper expands from increased temperature due to I^2R losses, the insulation, which is glued to the copper and wedged tightly between the conductor and the core, cannot expand due to a lower coefficient of thermal expansion and lower temperature. Repetitive stresses from sudden load changes strain the mechanical bond between the groundwall and either the strand or turn insulation, causing the bond to eventually weaken and break. This is mostly observed on long-core machines, that is machines with coils or stator bars longer than 2 m (6 ft). Gas turbine generators and pumped storage units, due to their nature of operation, are more susceptible to this type of insulation deterioration. Motors with long stator cores probably suffer the worst damage from thermal cycling because of surges from repetitive starts and stops.

A void bounded by the copper conductor and insulation exhibits a different phenomenon than those within the bulk of the insulation. Though the basic breakdown mechanisms are the same, the electrodes are of dissimilar materials and so polarity predominance occurs. In this case, there will usually be an observable *predominance of negative PD* pulses clumped at 45° during the positive AC cycle, as shown in Figure 7 [1].

For machines with multi-turn coils, this pattern raises concern about the integrity of the turn insulation

As with internal delamination, there tends to be an inverse temperature effect. If the voids are large, it is possible for the negative PD to increase with load due to the higher conductor vibration associated with the higher mechanical forces.

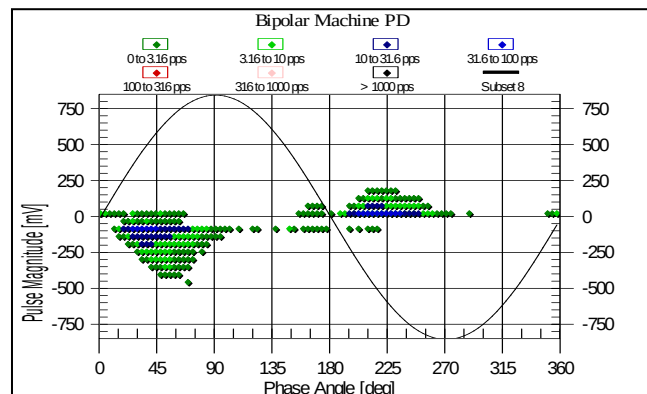


Figure 7: PD near Copper Conductor

Failure mechanism	Polarity	Load Effect	Temperature Effect	Phase Location
Thermal Cycling	Negative	None	Inverse	45°

4.2.2 Slot Partial Discharge

Loose coils and poor semi-conductive coatings can lead to discharge between the surface of the stator bar and the grounded core, called slot discharges. As with those near the copper conductors, these discharges occur between electrodes made of different materials. The observable PD pulses will be *predominantly positive and clumped at 225°*, as shown in Figure 8 [1]

4.2.2.1 Bar and Coil Movement

If properly installed, the wedges and side packing should prevent winding looseness. However, some insulation resins shrink when they are cured or thermally aged, coils may get smaller and so become loose in the slot. Also, some of the wedging and packing materials may become brittle and shrink over time, allowing the coils to become loose. In the presence of oil, side packing and ripple springs will soften faster because of the lubricating medium.

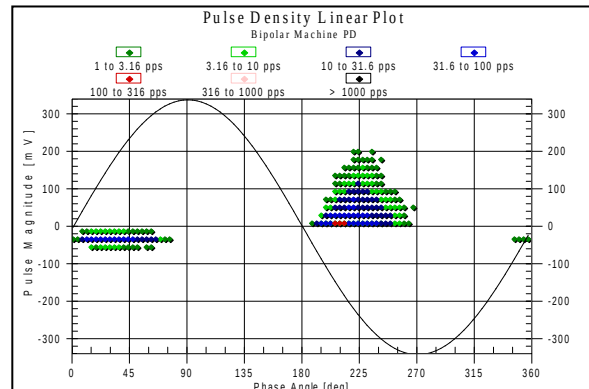


Figure 8: Slot-based Surface Activity

When windings become loose in the slot, the immediate problem is that, if left unattended, the looseness and vibration will quickly allow the laminated rough stator core surface to damage the semi-conductive coating on the surface of the coils. Damaged coil surfaces create discontinuities on the surface and allow voltage stresses to build up across these isolated locations, or between these and the stator core. If the voltage stress exceeds the electrical breakdown point of the gas medium, a discharge will occur. Eventually, a so called visual “ladder effect” develops where the groundwall insulation is worn out at the point of contact with the core, but maintains normal thickness at the core ventilation duct positions. Though the absolute time between the detection of looseness and failure is unknown, it can be as short as two years in some thermoset (hard) windings, especially in those with a high electric stress across the groundwall.

Load dependent PD may be an indication of winding looseness, which can damage a winding fairly quickly

As a result, pulses that occur from loose windings are predominantly surface-type activity within the slot portion of the core and therefore *positive and centred near 225°*. The most obvious sign of bar or coil movement is a noticeable increase in PD activity, especially positive activity, with an increase in load from a no load to full load, or a *direct load effect* (Figure 9).

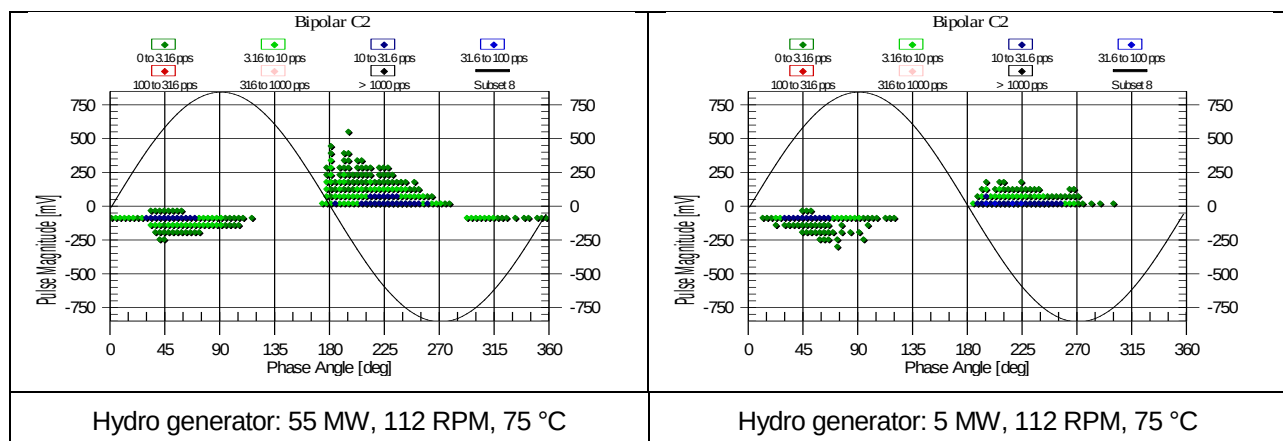


Figure 9: Direct Load Effect



Be aware, however, that the increase in load current will also increase the winding temperature, which may lead to a reduction in PD due to *inverse temperature* effects. Therefore, when checking for loose windings, run the PD test with the machine at different loads but at the same voltage, winding temperature, and hydrogen pressure. Note, for a winding with only a “looseness” problem, mainly the positive PD will change with load; the negative component will only change slightly, if at all. Changes in the negative PD with changes in load may be an indication of conductor vibration due to significant voids near the conductors.

Failure mechanism	Polarity	Load Effect	Temperature Effect	Phase Location
Bar/Coil Movement	Positive	Direct	Inverse	225°

4.2.2.2 Electrical Slot Discharge

Slot discharge is the term given to discharges that occur between the surface of the coil and the stator core. It can be said that PD caused by a loose winding creates slot discharges, but not all slot discharges are due to winding movement. Some of them are due to problems with the semi-conductive coating, and called electrical slot discharges. If the semi-conductive layer on the coil surface deteriorates, it results in the development of electrical slot discharge (a form of PD) and the production of ozone. Both PD and ozone will accelerate the decomposition of the organic epoxy or polyester resin binders. Due to this decomposition of the resins, the coils will decrease in size and may become loose. If allowed to become loose, the semi-conductive surfaces of the coils are the first to be damaged causing areas of high electric stress. If the semi-conductive coating is poorly made, especially if paint was used, it may become non-conductive at high temperatures and result in high electric stresses. However, as long as the coils remain tight in the slots, electrical breakdown to failure may take decades. Thus, machines with slight damage to the coil surfaces can still provide many years of reliable operation provided movement is stopped by re-wedging. Once damaged, it is difficult to restore the semi-conducting surfaces - so prevention is paramount. Windings with severely damaged surface coatings may require replacement because of the high production of ozone. Slot discharge occurs primarily in air-cooled machines.

Deterioration from surface PD sources is usually a slowly developing and long-term ageing mechanism, unless accompanied by a direct load effect (see Section 4.2.2.1), which usually indicates coil/bar movement and is especially a problem for thermoset windings.

The PD pattern for electrical slot discharge is very similar to coil/bar movement with *positive PD* near 225° predominating and an *direct temperature effect*. Since electrical slot discharge can occur before windings become loose, there is no noticeable load effect for electrical slot discharge. Electrical slot discharge can be very intense and in fact be a source of damage itself, for example, it can cause core damage.

Failure mechanism	Polarity	Load Effect	Temperature Effect	Phase Location
Electrical Slot Discharge	Positive	None	Direct	225°

4.3 Discharges in the Endwinding

Several PD sources may be found outside of the slots in a rotating machine. Often the presence of these discharge sources can be related to dust contamination, inadequate spacing and/or poor contact between the semi-conductive and the field-graded areas. Most of the aforementioned phenomena give rise to surface discharges, as opposed to discharges internal to the insulation, and may eventually lead to a phase-to-phase or phase-to-ground failure. Discharge activity on the circuit ring bus or between phase leads is also usually phase-to-phase voltage dependent discharge activity.

4.3.1 Surface Discharge at Voltage Stress Control Coatings Interface

To adequately control the voltage stress of a winding, the interface of the semi-conductive and grading coating materials must be effective [Figure 10]. Due to high electric stresses and temperatures, this interface may deteriorate over time, especially when paints are used for voltage stress control. As a result of this deterioration, the grading coating loses ground contact, floats to a high-voltage and sparks across the interface to ground. In air-cooled machines, this will produce ozone and leave a white band of residue near the slot exits. The band is readily seen when machines are inspected. Insulation damage from this type of deterioration is very slow as the PD takes place on the surface of the coil.

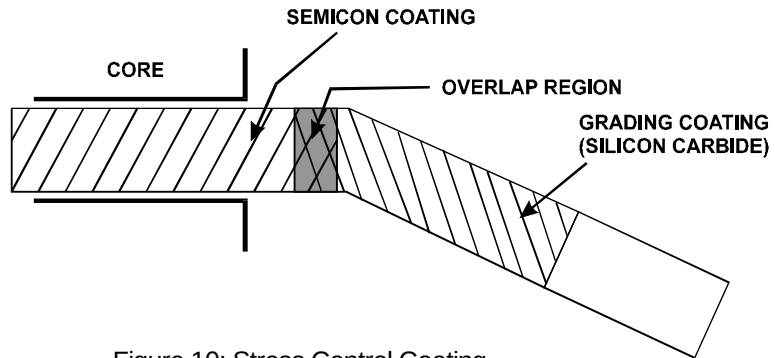


Figure 10: Stress Control Coating

Deterioration from the voltage stress coatings is usually a slowly developing ageing mechanism, but can produce high volumes of ozone in air-cooled machines

PD that occurs at the semi-conductive and grading coatings is a surface-type activity that is phase-to-ground voltage dependent. The activity is predominantly *positive* and centered *near 225°*. Unlike loose windings, this activity is not usually affected by load changes. An increase in temperature is usually found to lead to an increase in the PD activity resulting from this area, or a *direct temperature effect*. If a PD pattern is predominantly positive, centred near 225°, and increases with temperature, it is most likely from the semi-conductive and grading coating interface deterioration.

Failure mechanism	Polarity	Load Effect	Temperature Effect	Phase Location
Interface problems	Positive	None	Direct	225°

A severely deteriorated interface or arcing directly at the slot exit may lead to pulses clumped at 0° and 180° sometimes exhibiting a hook pattern (sometimes referred to as “rabbit ears” pattern) as it tracks the AC cycle (Figure 11) and other times simply vertical spikes (Figure 12).

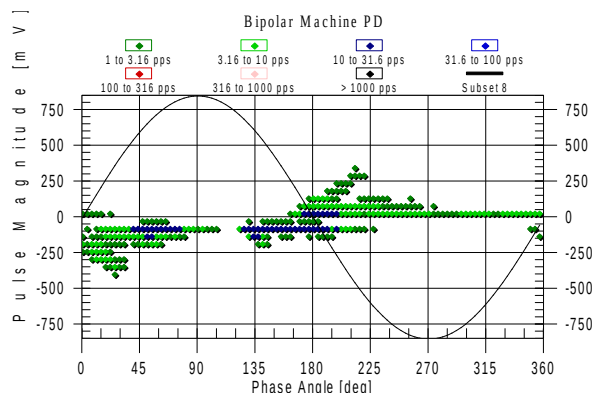


Figure 11: PD at Voltage Stress Coatings (“Rabbit ears”)

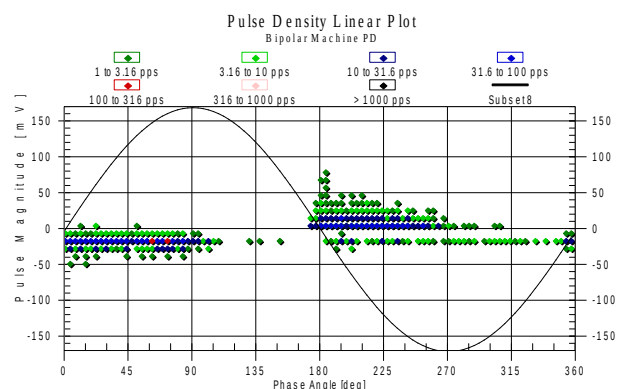


Figure 12: PD at Voltage Stress Coatings (Spikes)

4.3.2 Phase-to-Phase PD

Contamination and/or Inadequate Spacing in the end arm area, on circuit rings or on leads can produce partial discharge activity in these areas. In order to reduce the size of the coils and to save copper or reduce losses, manufacturers occasionally fail to leave adequate clearance between the coils in the endwinding area or the ring bus connections. If two adjacent components from different phases do not have sufficient spacing between them, it is highly likely that partial discharge activity will occur between the two. In air-cooled machines, this leaves a white powder residue. The discharges will slowly erode the groundwall and eventually puncture it. The closer the coils or components are – the faster the failure is. Generally these phase-to-phase faults take several years to develop, but they produce high quantities of ozone in air-cooled machines. However, if this activity is occurring between jacketed cable leads in the machine main terminal box, it can cause rapid insulation failure since such cable insulation has a lower PD withstand capability. The combination of inadequate spacing and a polluted operating environment can provide a fertile condition for PD activity.

4.3.2.1 Contamination

When any kind of conductive contamination from moisture or oil mixed with dust/dirt pollutes a machine, it is possible for electrical tracking (treeing) to develop across the blocking or along the end arms. Electrical tracking occurs because the pollution introduces a conductive path between two adjacent coils of different potential, and often from different phases. As a result of the tracking, PD can arise and attack the groundwall insulation on the surface. Permanent phase-to-phase or phase-to-ground deterioration may occur and eventual failure of the groundwall is possible. Moisture condensation in the slot section can lead to problems if a machine has been idle. Oil tends to dissolve and loosen insulation system components and can attract dust that diminishes heat transfer from the winding surface thus reducing insulation life. In open enclosure machines, oil, in combination with dust, can clog up cooling air passageways to cause overheating. Conductive particulate contamination could produce a strong local concentration of partial discharges that could produce small perforations in the insulation. [12]

Since the endwinding area of machines is specifically designed to be away from ground planes, PD that occurs due to contamination in the endwinding is normally phase-to-phase voltage dependent and will be 30° phase shifted from the positions of phase-to-ground PD of 45° and 225°. The direction of the shift is dependent on the location of the contamination and the voltages of the affected coils and is therefore unpredictable.

Identification or inspection of the site of arcing may be difficult due to limited visibility in the area. In situations of widespread contamination, the PD patterns may be more complex including patterns of phase-to-ground, phase-to-phase and open-air arcing as shown in Figure 13. Though there may be some effect of temperature or load changes on PD activity due to contamination, it is difficult to predict the expected results. However, environmental conditions such as humidity may have a noticeable effect, especially if the surface contamination becomes partly conductive when damp.

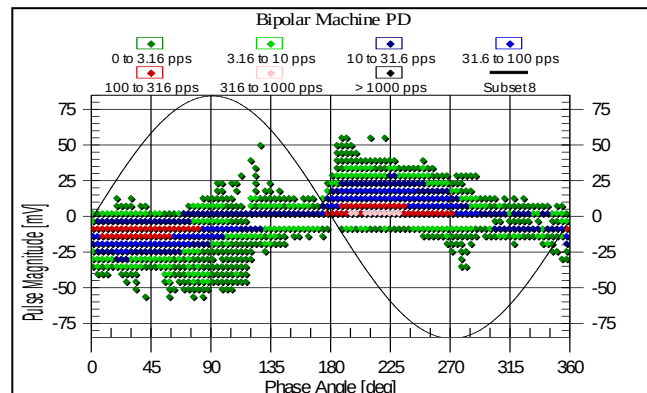


Figure 13: Mixed phase-to-ground and phase-to-phase discharges

Failure mechanism	Polarity	Load Effect	Temperature Effect	Phase Location
Contamination	No predominance	None	Unpredictable	15°, 75°, 195°, 255° (Can be erratic)

4.3.2.2 Inadequate Spacing

Electrical field distribution controls partial discharge. Since the fields in the endwinding area are phase-to-phase voltage dependent and the electrical fields in the slot area are phase-to-ground voltage, it is often possible to distinguish between discharge activities occurring in the slot part versus the endwinding part of the winding using Phase Resolved Plots [see Section 5.2.2]. Phase-to-phase voltage dependent PD pulses will be $\pm 30^\circ$ phase shifted from the positions of phase-to-ground PD, and centred at 15° , 75° , 195° and 255° positions along the reference sine wave, based on the phase rotation of the machine. Phase-to-phase activity is only confirmed by these pulse positions if the corresponding pulse patterns in the two phases at the same time are approximately equal in magnitudes and also have opposite polarities, as shown in Figure 14. For example, PD in one phase will be centred on 75° while the PD in the other phase must be centred on 195° of the AC cycle.

Although these pulses tend to be more erratic, it is sometimes possible to distinguish these pulses from others by observing their location with reference to the phase-to-ground voltage. Since this activity is frequently isolated to a specific area of the machine, the PD pattern may reveal which phases are involved. This is only possible if the two affected phases both have detectable endwinding activity and their respective 30° phase shifts correspond to what would be expected according to the machine rotation (Figure 14). Sometimes this PD may be occurring between a top and bottom coil in a slot, and visual verification of the activity may be difficult.

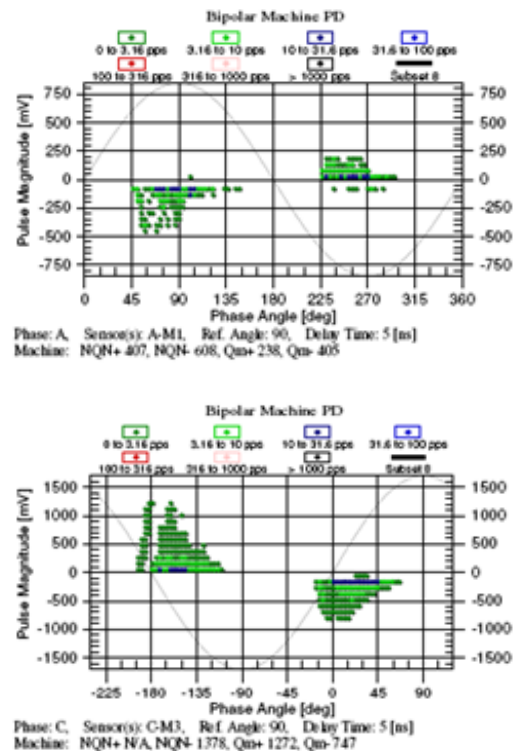


Figure 14: Phase-to-phase Discharges

Note that the origin of the AC cycle is different for the two phases.

Failure mechanism	Polarity	Load Effect	Temperature Effect	Phase Location	Pulse Polarity
Phase to Phase Discharges	No predominance	None	Unpredictable	15° , 75° , 195° , 255°	Opposite in the two phases

4.3.2.3 Gap-type Discharges

A gap-type discharge can occur when there is a constant space between two components at different potential. This type of activity will occur between two bars in the winding overhang, or between a bar and the pressure finger of the stator core. The resultant PD pattern is usually a “cloud”, that is, a cluster that occurs at constant magnitudes without any lower magnitude pulses as shown in Figure 15. The “cloud” may be clustered around the positions associated with phase-to-phase or the phase-to-ground sources according to the location of the gap-type discharge location.

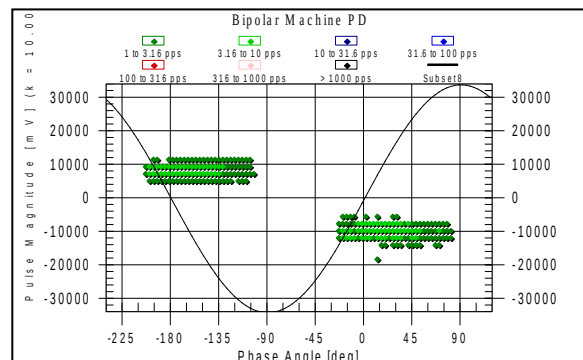


Figure 15: Gap-type Phase-to-phase

Failure mechanism	Polarity	Load Effect	Temperature Effect	Phase Location
Gap-type Discharges	No predominance	None	Unpredictable	Variable ("Cloud")

4.4 Other Arcing and Sparking

4.4.1 Broken Strands Arcing

When excessive endwinding vibration develops or due to manufacturing defects, some machines can experience broken copper conductors or strands of the stator winding. Such breaks tend to occur near the outer ends of the stator bars. Violent arcing occurs at the point of failure. This is not voltage driven PD but rather contact sparking with intensity of the activity usually dependent on a machine load, that is, current through the conductor. Similar arcing can occur at the bar jumper connections when there is a crack in the jumper or the brazed connection is unreliable. Although such arcing may produce plenty of high frequency activity in some cases, the detection of these PD-like pulses could be complicated due to remoteness of the source from PD coupler locations – the fault can be anywhere in the winding and even at the neutral ends. Sometimes useful information can be seen from the "System Activity" data plots. Please refer to **Zone of coverage**, Section 5.3

Failure mechanism	Polarity	Load Effect	Temperature Effect	Phase Location
Broken conductor arcing	Unknown	Direct	Unknown	Often at 0°/180° but can be erratic

4.4.2 Vibration Sparking

Although not common for all machines this deterioration (also known as spark erosion) can occur in some machine designs [13]. At least two conditions should be met for this to happen – the stator coil or bar would experience some movement (not necessarily great) and the slot semi-conductive coating is too conductive.

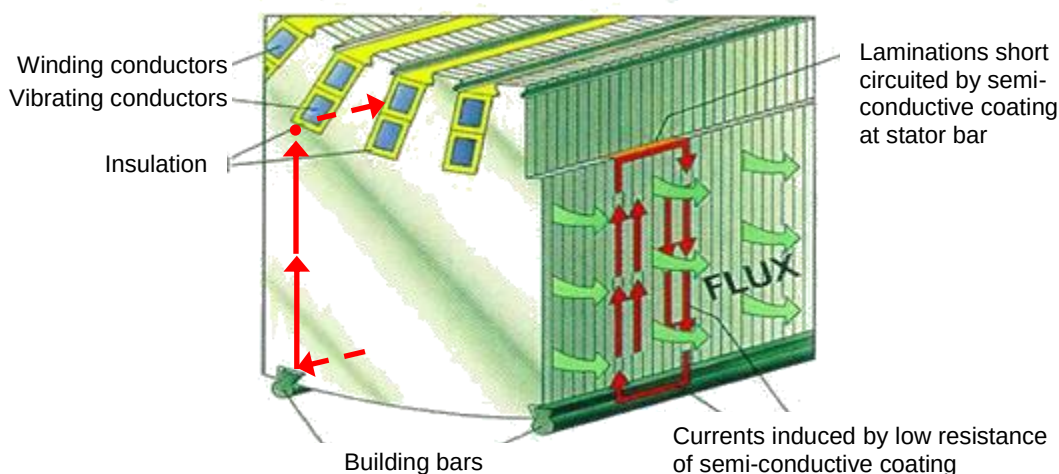


Figure 16: Induced Current from Vibration Sparking

In this case, contact sparking occurs when the moving coil or bar loses contact with the side of the stator core slot.

For lower resistance of the semi-conductive coating, the induced current as shown in Figure 16 is of such significance that the resulting sparks with the help of ozone can lead to accelerated degradation of the coating. This, in turn, will induce slot discharge and the affected machine can eventually fail only after a few years of operation.

Since the physics of the vibration sparking is not voltage driven, such contact sparking can occur at the low voltage end of the winding and the resulting PD pattern may not be locked to the AC sine wave. For this reason, early stage detection is complicated though eventually slot discharge should be reliably recognized as in 4.2.2.2.

Failure mechanism	Polarity	Load Effect	Temperature Effect	Phase Location
Vibration Sparking	Unknown	Direct	Inverse	Unknown

4.5 Extraneous Signals

As described in Section 3.5, “Disturbances are electrical pulses of relatively short duration that may have many of the characteristics of stator winding PD pulses – but in fact are not stator winding PD” [12]. Some of these disturbances are synchronized to the AC cycle, and some are not. Sometimes synchronized disturbance pulses can be suppressed based on their position with respect to the AC cycle. See Section 5.1.2 for more information about Disturbance separation.

Hence, as part of the data analysis process, it is important to identify whether the pulse patterns displayed contain disturbances from sources within the stator winding environment that could not be separated by the PD instrument or test configuration. Disturbance-like sources tend to produce patterns that occur across the AC cycle. There are three common disturbance patterns that may occur.

4.5.1 Excitation pulses

PD couplers sometimes detect the high frequency thyristor-firings of static excitation systems. These pulses are usually of equal pulse repetitions, low magnitudes and occur repetitively across the AC cycle (Figure 17). Usually, they are only discernible when there is low PD activity on the unit and the test range of the PD test instrument is set to its most sensitive (lowest range) setting. This is not usually indicative of a problem with a stator winding, but if pulse levels are high this may indicate that there is a problem in the exciter. Although this type of pattern is called “excitation pulses”, similar signals may also be detected from machines close to variable frequency drives in the electrical system.

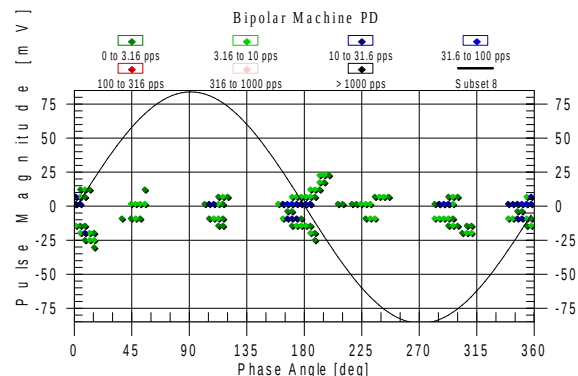


Figure 17: Excitation Pulses

Disturbance pulses are not indicative of PD sources within the winding insulation.

4.5.2 Cross-coupling

PD pulses are high frequency pulses that may capacitively couple from one phase to another across the endwinding or circuit rings. In addition, a coupler connected to one phase may detect pulses radiated from a source on another phase. This behaviour is called cross-coupling and can be detected by observing PD pulses shifted $\pm 120^\circ$ from the positions of the source PD. Phase-to-ground PD cross-coupled from another phase would cluster at 345° , 105° , 165° or 285° . Cross-coupled inter-phase activity would be $\pm 30^\circ$ phase-shifted from these positions as shown for C-phase (middle plot at 315°) in Figure 28.

Cross-coupled PD is dependent on the geometry of the stator winding and does not necessarily indicate a problem with the insulation. Additionally, based on the mode of pulse travel, the pulses may invert in polarity.

**Cross-coupled pulses are not indicative of PD sources,
and therefore, are not usually associated with ageing mechanisms.**

4.5.3 Broadband Clusters

Due to the physical attributes of some installations, there may be sources of low-magnitude broadband clusters detected by the couplers. When this situation exists, the pulses are low in magnitude and normally manifest themselves across the AC cycle (Figure 18). In some instances, the magnitude of the detected pattern could be reduced when a more reliable (or shorter) ground path for the PD couplers was chosen.

Typically the magnitude of this activity is quite low and as the winding ages, PD originating within the winding will reach higher magnitudes and be visible beyond this noise-like disturbance.

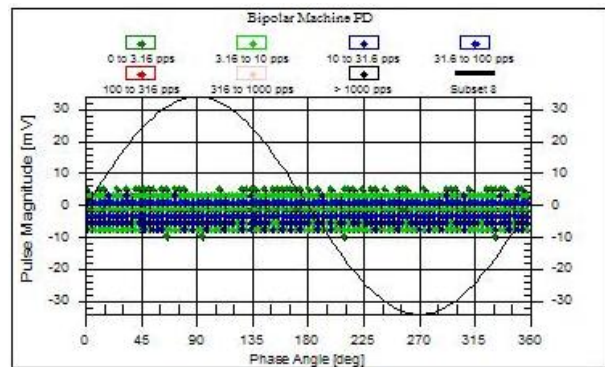


Figure 18: Broadband Clusters

**Broadband clusters are not indicative of PD sources
within the winding or problems with the PD detection system.**

4.5.4 High level disturbance

Sometimes, sources external to the machine can produce such a high level of electromagnetic energy and induce disturbances in the stator winding that exceed the typical PD activity by orders of magnitude. One typical example would be an air gap between the bus-bar and the bus-bar support. The intense sparking at this location will induce high frequency pulses that will travel to the PD couplers and be detected there. If such external disturbances occur close to the machine, they may be classified as PD. Measurements in low (for PD) and high (for disturbance) sensitivity ranges may be required to distinguish between the two. Additional instrument counters (Between Activity, for example) can also assist in these complicated cases. Please note that although external disturbance is not in the scope of machine PD monitoring, high external discharge could be associated with malfunctions of connected auxiliary equipment and it could lead to a ground fault.

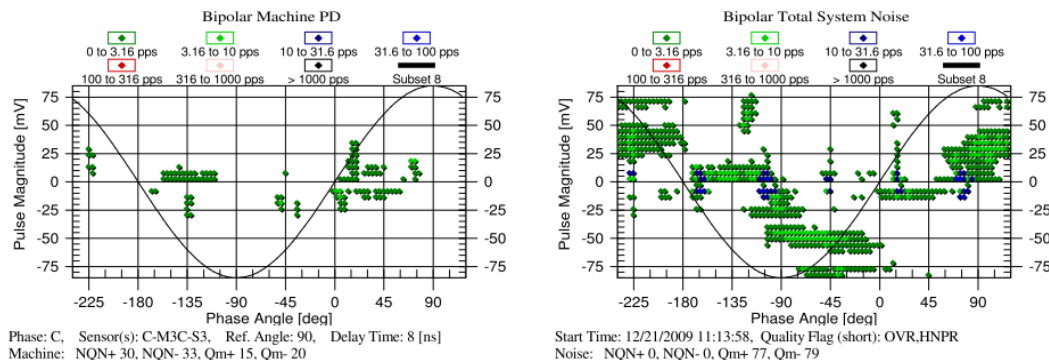


Figure 19: External Disturbance in Machine PD Plot That Is Clearly Not Winding PD



4.6 Summary - Failure Processes

Failure mechanism	Polarity	Load Effect	Temperature Effect	Phase Location
Internal delamination	No predominance	None	Inverse	45° and 225°
Thermal cycling	Negative	None	Inverse	45°
Manufacturing defect	No or negative predominance	None	Inverse	45° and 225°
Bar/coil movements	Positive	Direct	Inverse	225°
Electrical slot discharge	Positive	None	Direct	225°
Contamination or Electrical tracking	No predominance	None	Unpredictable	15°, 75°, 195°, 255°, can be erratic
Interface problem	Mostly positive	None	Direct	225°, can be 0°/180°
Gap-type discharge (Open air)	No predominance	Unknown	Unknown	0° - 90°, 180° - 270°
Vibration sparking *	No predominance	Direct	Inverse	Erratic
Broken strands *	No predominance	Direct	Unknown	Sometimes 0°/180°, can be erratic
* Will only be seen in PD measurements if it is occurring at or near the phase line ends or it is near the neutral connection and an EMC is installed there.				



5 APPENDIX

5.1 Noise and Disturbances

5.1.1 Noise Suppression

Noise is defined to be “non-stator winding signals that clearly are not pulses” [12]. Electrical noise from radio frequency sources, excitors and power supplies etc., may be confused with PD from the machine windings. Noise may be due to electronic devices within the PD detection system itself.

One of the most effective ways of minimizing *noise* is frequency domain separation, or to test at frequencies high enough that only the higher frequency, over 50 MHz, partial discharge pulses will be detected and the lower frequency, below 40 MHz, electrical noise pulses will be suppressed. The key is to select an appropriate frequency range such that the signal-to-noise ratio (SNR) is so high that only PD signals and not noise are counted.

The sensing device defines the low-end frequency limit. The 80 pF sensors terminated into 50 Ω impedance limit the lower cutoff frequency of detection to 40 MHz. Above this frequency, the noise signal is normally eliminated and therefore the Signal-to-Noise-Ratio is high. Limiting the sensor to detect only high frequency signals reduces the total amount of PD energy found, but more importantly, it also eliminates the need for an expert to discriminate between noise and PD. This creates a test that a non-expert can perform and data analyzed.

The upper cutoff frequency is set by the test instrument. The test instrument has an input bandwidth of 0.1-350 MHz. It is capable of detecting the fast rise-time initial pulses and the high frequencies (50-250 MHz) typical of PD originating in the winding. Using the 40 MHz cut-off frequency of the sensor and the 350 MHz upper frequency limitation of the instrument, the frequency band of 40-350 MHz is monitored. As shown in Figure 20, this enables the detection of the fast rise-time PD pulses without noise interference by *maximizing the signal-to-noise ratio*.

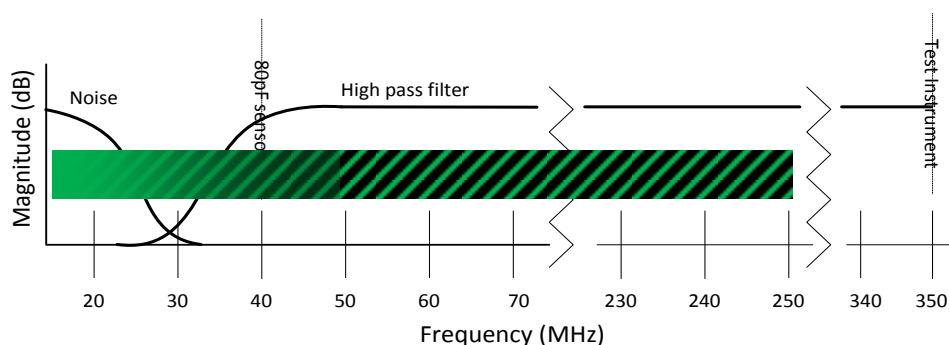


Figure 20: Frequency Characteristics

High frequency signal pulses attenuate fairly quickly within the winding so only pulses occurring across voids in coils near the high voltage end of the winding will be detected with high frequency testing, i.e., above 40 MHz. However, as stated before, pulses across voids are highly dependent upon the applied voltage to ground, thus the coils near the high voltage end are the ones likely to have PD. Since most failures occur in these coils, and the common failure mechanisms affect the entire winding, high frequency testing is quite successful in detecting most failure mechanisms.

5.1.2 Disturbance Separation

“Disturbances” are electrical pulses of relatively short duration that may have many of the characteristics of stator winding PD pulses – but in fact are not stator winding PD [12]. Examples include corona from the power system and sparking from poor contacts etc.

5.1.2.1 Impedance Mismatch

One method of disturbance separation is to use the natural tendency of a high frequency pulse to distort when it traverses from a conductor of one surge (characteristic) impedance level to another, or *impedance mismatch*. The effect is a change in the magnitude (attenuation or amplification) and the rise-time (frequency) of the pulse. Since the typical surge impedance of a circuit ring is about 100 Ω , and that of a winding is of the order of 30 Ω , the magnitude of a high-frequency pulse traveling across this junction will be altered.

A pulse that originates in the winding will almost double in magnitude as it travels from a low impedance coil to a high impedance circuit ring. Meanwhile, pulses from the system (disturbances) will nearly halve in magnitude travelling from a high impedance circuit ring to a low impedance coil. Because of this phenomenon, the high voltage leads of the sensing capacitors should be attached to the winding as close as possible (less than 1 metre/3 feet) to the junction of the first coil and the circuit ring. Sensors in this position will have the maximum sensitivity to machine PD and the maximum attenuation of disturbances.

5.1.2.2 Time Domain Separation

Perhaps the most effective means of separating PD-frequency disturbances is via the time-of-arrival method. Though this method does require extra effort during installation, it ensures that PD pulses from outside the machine are classified as disturbances (previously called noise). Two different installation configurations can be used for time-of-arrival separation: differential and directional.

5.1.2.2.1 Differential (PDA)

In hydro-generators, coupler pairs are placed on each phase and the length of the coaxial cables are adjusted in such a way that any system disturbance pulses detected by the two couplers will arrive at the test instrument simultaneously (Figure 21). The test instrument will perform a differential comparison of arrival time along with the pulse shapes, sizes and polarities, thereby separating disturbance from machine PD.

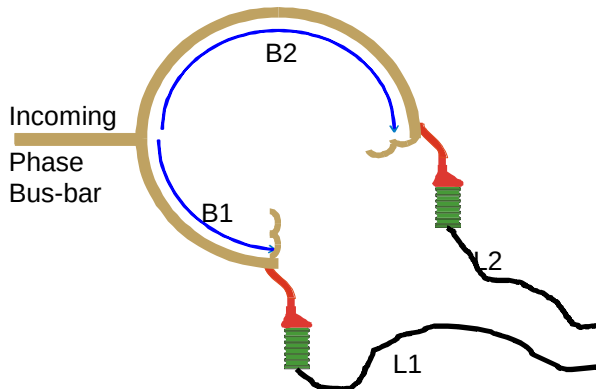


Figure 21: Differential Disturbance Separation

During calibration, the coaxial cables are adjusted so that: $B1 + L1 = B2 + L2$ (in nanoseconds)

These represent the travel time for a high frequency pulse.

Disturbance pulses originating in the power system arrive at the same time at the end of the coaxial cables connected to each coupler, and will be classified as disturbances. All other pulses (PD) will be assigned to the coupler that detects them first, and will be classified as PD. This is a *differential time-of-arrival* disturbance separation.

The optimum coupler placements in a generator are dependent on the layout of the stator winding. If certain physical space is not available, it may not be possible to install pairs of couplers in places that best separate the effects of external disturbance. The preferred differential style of coupler installation can be done on hydro-generators that have at least 1 metre (3 feet) of circuit ring bus on each of the parallels to be monitored, for a total distance between couplers of at least 2 metres (6 feet). For machines with insufficient circuit ring length, a PDA directional installation is recommended.

For machines smaller than 100 MW, two couplers per phase are generally installed, one each at the line end of different stator parallel circuits. If a generator has more than two parallels per phase, it is possible to have a coupler on each parallel for more coverage. This is customarily reserved for larger units (such as over 100 MW) or for very important smaller units where the extra cost is justified. When carrying out the calibration for machines with more than two couplers per phase, couplers are “paired” for calibration and data collection.



5.1.2.2.2 Directional (BUS)

If there is no circuit ring, or it is less than 2 metres (6 feet) long (as required for the differential style), a directional installation is necessary. For a directional installation, one coupler (Machine coupler) is placed as close as possible to the junction between the line-end coil and the circuit ring. The second coupler (System coupler) is connected to the phase bus-bar, at a convenient location towards the system and at least 2 metres (6 feet) from the first coupler. To use the TGA-B instrument, the coaxial cables should be the same length (Figure 22).

Calibration involves measuring the delay time or the time a fast rise-time pulse takes to travel between the couplers. In a directional installation, instead of the system disturbance arriving at the couplers from opposite directions, it arrives from the same direction at the end of both coaxial cables. Pulses arriving at the Machine coupler first are classified as Machine PD, pulses arriving at the System coupler first are classified as System Activity (previously called System Noise), while those arriving at both sensors within the Delay time are called Between Activity (previously Between Noise).

“L1” represents the time in nanoseconds that a pulse takes to travel through the coaxial cables. For a directional installation, the coaxial cables are the same lengths; so L1 is equal for both couplers. “Delay” is the time in nanoseconds for a pulse to travel along the bus-bar between the two couplers.

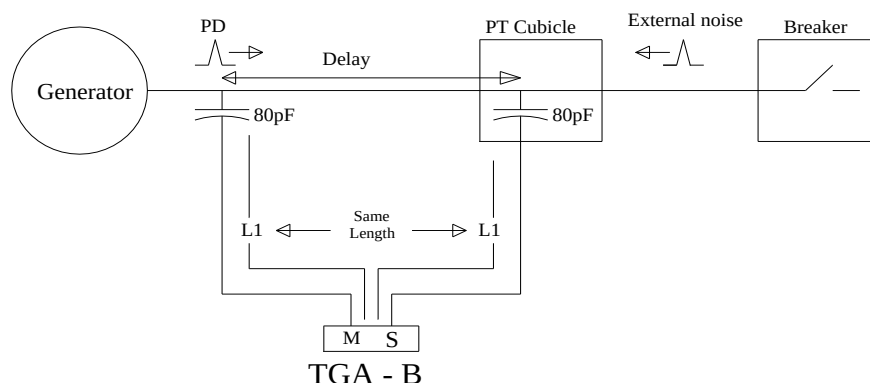


Figure 22: Directional Time-of-Arrival Disturbance Separation

5.1.2.2.3 Directional (SSC)

Like the Directional (BUS) installation, the SSC installation has two sensors, one on each end of the antenna. It categorizes pulses by the direction of arrival. The delay time between pulse arrivals is part of the sensor design and does not require calibration during installation. Pulses arriving at the “Endwinding” sensor first are classified as “Endwinding PD”, pulses arriving at the “Slot” sensor first are classified as “Slot PD”, while those arriving at both sensors within the Delay time are called “Under PD”.

5.1.2.3 Pulse Shape

5.1.2.3.1 SSC

Experiments show that internal disturbances when detected by an SSC have a pulse width longer than 8 ns and external disturbances have a pulse width more than 20 ns, whereas PD in the slot has a pulse width shorter than 6 ns. The TGA-S instrument, in addition to the directional signal separation described above, measures the width of each pulse and classifies it as PD, disturbance or wide pulses.

5.1.2.3.2 Single-ended (BUS)

In some installations, particularly motors and smaller generators, the unit is connected to the power system with a relatively long power cable. If the power cable is longer than about 30 metres (about 100 feet), the high frequency disturbances from the power system may be severely dispersed and attenuated when detected at the machine terminals. Thus, if a pulse observed at the machine terminals has a fast rise time it is probably PD; whereas if it has a long rise time, it is likely a disturbance. In these cases, only one coupler per phase, or three per unit, is necessary. This is often referred to as a “single-ended installation”. There is no calibration required, though a sensitivity check is recommended.

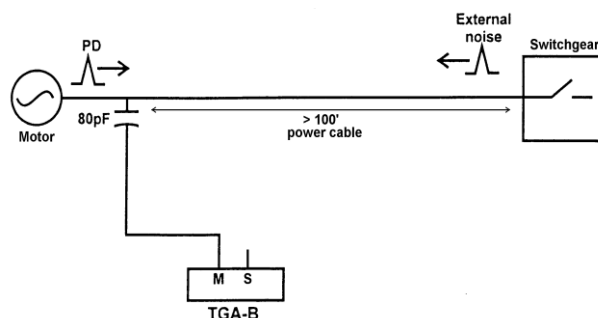


Figure 23: Single-ended Installation

This configuration is used for PDTrac or PDTracII monitors, but can also be used with a TGA-B or PDA-IV portable instrument.

5.2 Data Types

5.2.1 PD Summary Variables

Pulse Height Analysis (PHA or 2D) Plots represents the relationship between the pulse magnitude and the pulse repetition rate for both negative and positive polarity PD pulse counters, for instance Machine +PD and Machine -PD as shown in Figure 24. The horizontal axis is a linear representation of the pulse magnitudes while the vertical axis is a logarithmic representation of the number of pulses per second. Iris instruments can be operated on several input sensitivity test ranges and using two different time durations for each magnitude window within a range: 1 second and 5 seconds. Pulses are always displayed on a “per second” basis. The greater the number of pulses per second, the more widespread is the deterioration in the winding. The higher the PD pulse magnitudes, the more severe the deterioration. Only negative pulses that occur from -15° to 135° and positive pulses from 165° to 315° relative to the power frequency voltage waveform are displayed in the PHA plot.

Iris instruments calculate the quantities Q_m and NQN based on the entire Pulse Height Analysis (PHA) plot. The NQN, or *Normalized Quantity Number*, is a partial discharge quantity that is proportional to the total partial discharge measured by a PD sensor. The negative NQN refers to the total activity from negative PD pulses, while positive NQN refers to the total PD activity from positive PD pulses. NQN is an indicator of the average condition of the stator winding insulation (Figure 24).

The Q_m , or *Peak Magnitude*, is the magnitude of the pulses for one fundamental (directly measured) pulse category that has a repetition rate of 10 pulses per second, and corresponds to the peak PD activity. Q_m is an indicator of how severe the PD is at the most deteriorated part of the winding. Positive and negative Q_m refers to the peak PD activity from the positive and negative PD pulses, respectively.

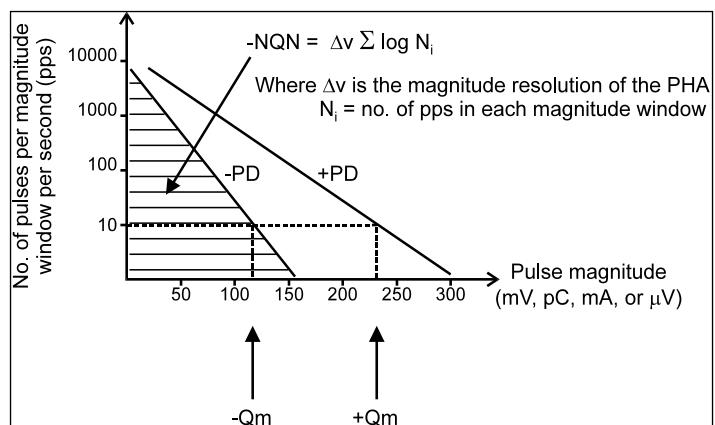


Figure 24: PD Summary Variables

For our purposes, Q_m and NQN are derived from the test results and are valuable for comparisons amongst machines or trending over time. The Trend Plot represents the trend in partial discharge summary numbers, specifically NQN and Q_m , and thus provides a graphical representation of the progression in insulation ageing. The Trend Plot provided in the PDView software may display trends of up to 10 parameters for a given selection of data sets, including $\pm$ NQN and $\pm Q_m$, load, and temperature (Figure 25).

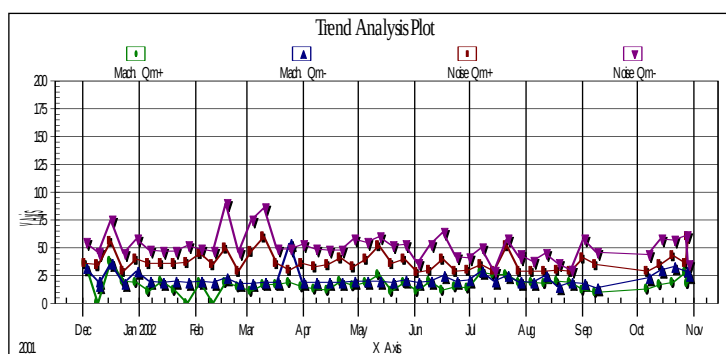


Figure 25: Trend Plot

5.2.2 Phase Angle Resolved Plots

During data collection using a Portable, Guard or GuardII instrument, each cycle is divided into 100 phase windows to enable the display of data relative to the specific phase-to-ground voltage reference, or 24 phase windows for a Trac or TracII monitor. Experience has indicated that such pulse phase analysis is useful to help identify what the deterioration mechanisms are and if multiple mechanisms are occurring.

The Pulse Phase Analysis (3D or PPA) or Linear Pulse Density Plot (LPD or referred to as PRPD by some in the industry [13]) is the most commonly used graph type for display of phase-resolved partial discharge data [Figure 26]. This two-dimensional plot displays the partial discharge magnitude (vertical axis) as a function of the phase angle of the reference 50/60 Hz sine wave (horizontal axis). A colour-coding scheme provides information on the pulse repetition rate for a given combination of phase-angle and magnitude.

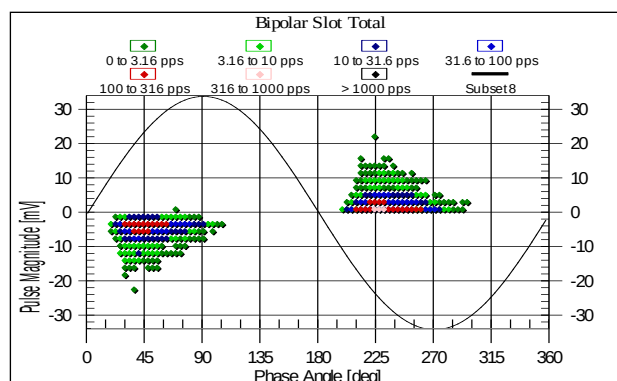


Figure 26: Linear Pulse Density Plot

In phase-resolved analysis, the reference voltage usually chosen for a given measurement is the phase-to-ground voltage of the phase subjected to on-line testing. Please note that the major unit division of the pulse phase scale (the x-axis) is 45° and the minor unit division is 15° . This allows for quick identification of the phase angle value at which a given PD cluster is centred. Specifically, this allows for quick discrimination between slot and inter-phase partial discharge sources [see Section 4]

The standard LPD report contains six graphs organized in three rows, one for each phase, and two columns, one per PD classification. The data can be displayed in Standard form, where all three phases are shown relative to their respective phase-to-ground voltages (

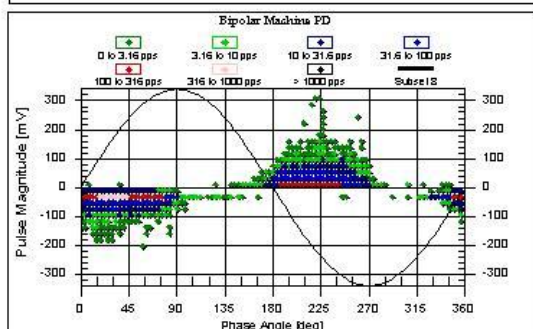
Figure 27); or in phase-shifted form, where the first row of graphs displays data measured on one phase with no phase shift, and the second row of graphs displays data measured on the second phase but with the phase-angle scale shifted 120° from the first phase. The third row of graphs displays data measured on the third phase shifted 240° from the first phase (Figure 28). Thus, the sinusoidal phase voltages are displayed as they occur simultaneously in time. This allows for quick identification of phase-to-phase voltage dependent discharge sources and cross-coupled PD clusters. In this configuration pulses from the same source, such as inter-phase arcing or cross-coupling will align vertically.

Pulse Phase Analysis Report

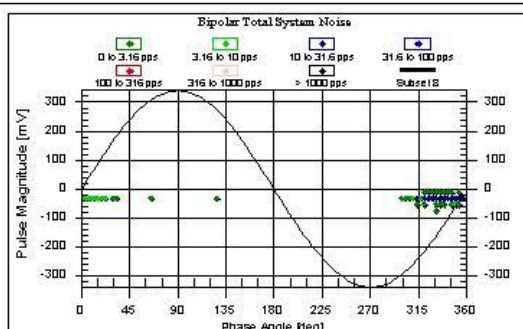


Organisation: N/A Asset Name: N/A
Asset Class: N/A Class: N/A

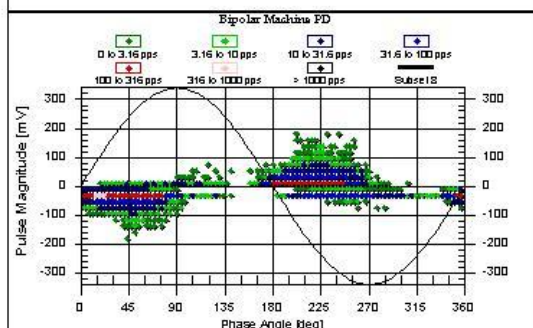
Load: 110.00 MW, Reactive Load: 42.00 MVar, Stator Voltage: 16.40 kV
Stator Temp: 0 deg C, H2 Pressure: 29.0 psi



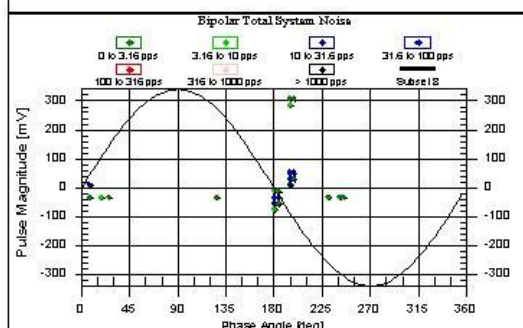
Phase: A, Sensor(s): M1, S1, Ref. Angle: 90
Mach.: NQN+0, NQN-401, Qm+185, Qm-157



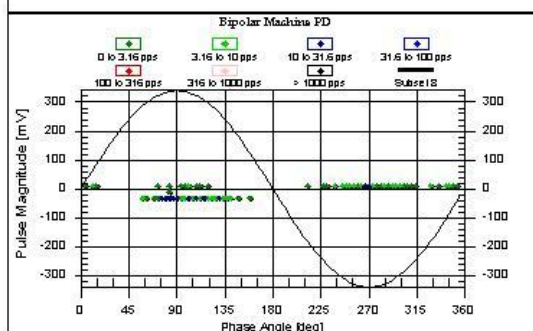
Delay Time: 10 [μs], Start Time: 07/02/1999 14:17:59
Noise: NQN+0, NQN-88, Qm+0, Qm-53



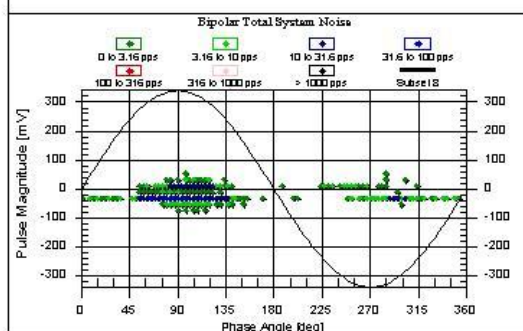
Phase: B, Sensor(s): M2, S2, Ref. Angle: 90
Mach.: NQN+398, NQN-320, Qm+173, Qm-141



Delay Time: 9 [μs], Start Time: 07/02/1999 14:18:32
Noise: NQN+0, NQN-26, Qm+77, Qm-28



Phase: C, Sensor(s): M3, S3, Ref. Angle: 90
Mach.: NQN+48, NQN-45, Qm+37, Qm-37



Delay Time: 12 [μs], Start Time: 07/02/1999 14:19:04
Noise: NQN+60, NQN-120, Qm+37, Qm-67

Iris Power Engineering, 2-1 Westside Drive, Toronto, ON Canada M9C 1B2, Phone: (416) 620-5600, Fax: (416) 620-1995

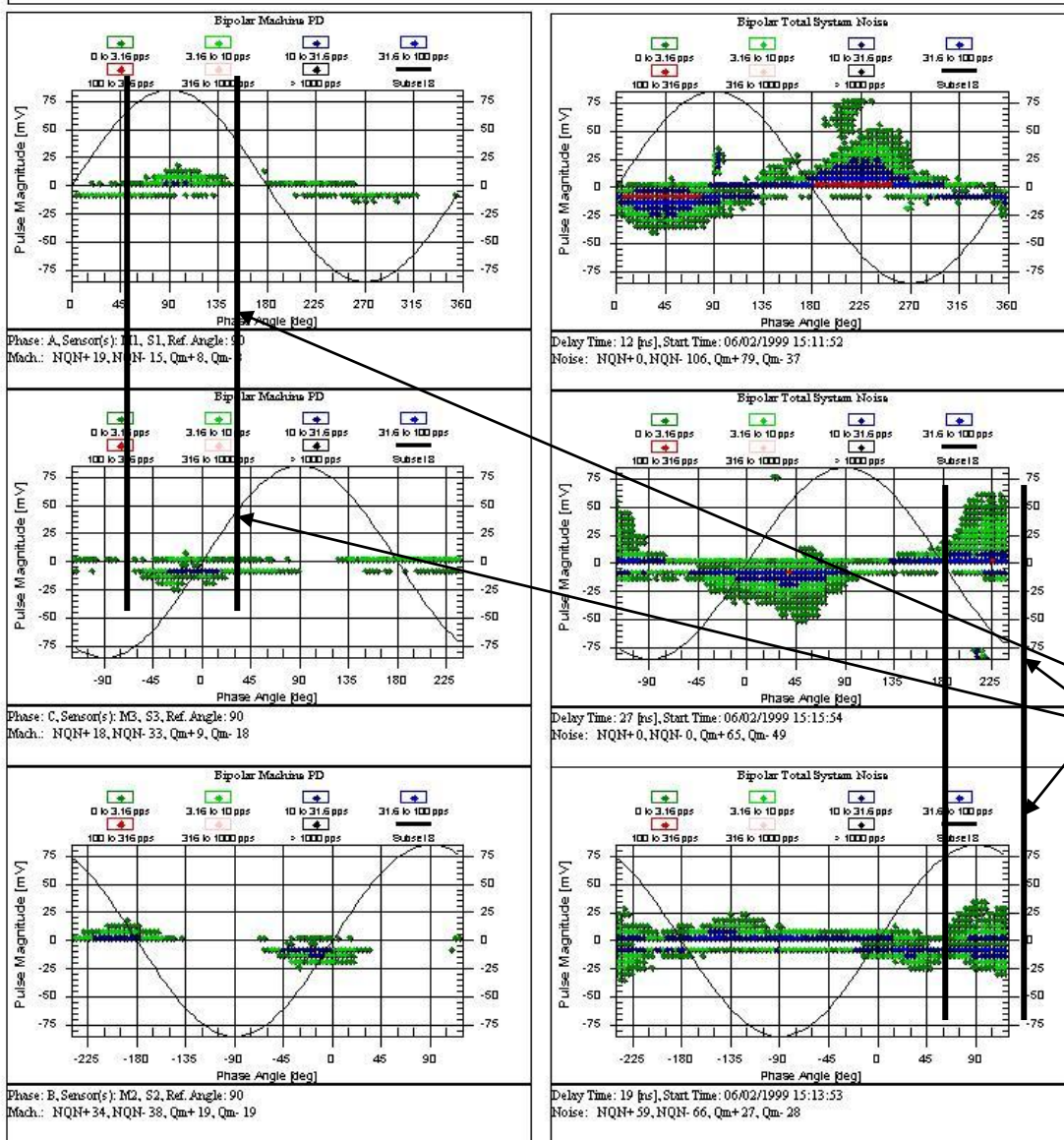
Figure 27: LPD Standard PD Report

Pulse Phase Analysis Report



Organisation: N/A Asset Name: N/A
Asset Class: N/A Class: N/A

Load: 283.00 MW, Reactive Load: 137.00 MVAr, Stator Voltage: 20.00 kV
Stator Temp: 123 deg C, H2 Pressure: 59.0 psi



Iris Power Engineering, 2-1 Westside Drive, Toronto, ON Canada M9C 1B2, Phone: (416) 620-5600, Fax: (416) 620-1995

Figure 28: Linear Pulse Density Plot - ACB Rotation PD Report



5.3 Zone of Coverage

5.3.1 TGA - SSC Test Method

(TGA-S, TurboGuard, GuardII)

This PD test requires the prior installation of PD sensors called stator slot couplers (SSC) within the stator winding. During normal machine operation, an instrument called the TGA-S is temporarily connected or a TurboGuard or a GuardII is continuously connected to each previously installed SSC.

These Iris instruments separate PD from system noise based on pulse characteristics, and measure the number, magnitude, location (endwinding, under or slot) and relative AC phase position of the PD pulses. The test in this configuration can reliably account for system noise and ensure there is no impact on PD measurements. It is the most reliable means of monitoring the PD activity in an operating turbine generator where internal machine disturbance can be a problem.

The TGA-S and TurboGuard or GuardII are capable of separating pulses into six categories:

- Slot PD – narrow pulses originating within the stator slot section of the bar
- Under PD – narrow pulses originating from underneath the SSC sensor
- Endwinding PD – narrow pulses originating from the endwinding section of the stator bar
- Slot Wide pulses – pulses between 8–20 ns in width originating within the slot section of the stator bar
- Endwinding Wide pulses – pulses between 8-20 ns in width originating in the endwinding section
- Noise – pulses wider than 20 ns or with non-PD like characteristics

5.3.2 TGA - BUS Test Method

(TGA-B, BusGuard, GuardII, BusTrac, BusTracII)

During normal machine operation, either the TGA-B is temporarily connected, or the BusGuard or GuardII, or BusTrac or BusTracII is continuously connected to the previously installed couplers in each phase [5]. These Iris instruments separate PD from System noise and disturbances based on pulse characteristics and time-of-arrival, while measuring the number, magnitude and relative AC phase position of the PD pulses. This test is a reliable means of monitoring the PD activity in operating turbine generators and motors, where internal machine “noise” is not generally a problem.

When two couplers per phase are installed, the TGA-B, BusGuard, GuardII, BusTrac and BusTracII have the capability of separating pulses into three categories:

- Machine PD – PD originating within the zone from the stator winding to the coupler, and about 1 m beyond
- System Activity – pulses originating away from the machine beyond the system coupler
- Between Activity– pulses originating between the machine and system couplers

The latter two are combined as Total System Activity for display purposes. The separation of the external disturbance pulses into “Between Activity” and “System Activity” can be viewed using PDView software and choosing an appropriate template. Note that the Zone of coverage for the Machine coupler includes pulses detected from the Machine sensor back into the winding; so if further investigations are warranted due to indications of high PD, they should always start at the machine terminals.



5.3.3 TGA - Single-ended Test Method

(TGA-B, PDTrac and PDTracII)

During normal machine operation, either the TGA-B is temporarily connected, or the PDTrac or PDTracII is continuously connected to the previously installed couplers in each phase [5]. If the machine being monitored has at least 30 metres (100 feet) of high-voltage shielded cable connecting the machine to the electrical power system or switchgear, only one coupler per phase is required, as the long high voltage cable will diminish external noise and disturbance interference.

In this configuration, the TGA-B has only one category, Machine PD, since there will be no capabilities for classification of disturbances by the time of flight method, while the PDTrac or PDTracII has two:

- Machine PD – all detected pulses with PD characteristics
- Noise – all pulses detected that do not exhibit PD characteristics

Note that the Zone of coverage for a single-ended installation will include everything; so if further investigations are warranted due to high PD, they should always start at the terminals.

5.3.4 PDA Test Method

(PDA-IV, HydroGuard, GuardII, HydroTrac and HydroTracII)

During normal machine operation, either the PDA-IV is temporarily connected, or the HydroGuard or GuardII, or HydroTrac or HydroTracII is continuously connected to the previously installed couplers in each phase. If there is an internal circuit ring of sufficient length, the best installation is a differential mode; otherwise, a directional mode is required.

Two or more couplers per phase are installed near the junction of the circuit ring and the high voltage coils of different parallels per phase [4]. These Iris instruments separate PD from System disturbances based on simultaneous time-of-arrival and pulse characteristics, while measuring the number, magnitude and AC phase position of the PD pulses. This configuration can reliably account for System activity and ensure there is no negative impact on unit PD measurements. It is a reliable means of monitoring the PD activity in a hydro generator where internal machine disturbance is not generally a problem.

The PDA-IV, HydroGuard, GuardII, HydroTrac and HydroTracII are capable of separating pulses into three categories:

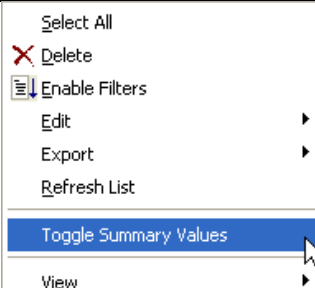
- C1 – narrow pulses originating near the C1 circuit
- C2 – narrow pulses originating near the C2 circuit
- System Activity – narrow pulses that arrive simultaneously at the two coupler outputs

Note that the Zone of coverage for each pair of couplers, C1 or C2, will include the portion of the circuit ring from where the coupler is connected back to where the circuit ring splits; so if further investigations are warranted due to high PD, they should always start at the junction of the circuit ring.

5.4 Prime (Alternate) Numbers

Because of the possibility of the reversal in polarity of phase-to-ground PD when using SSC's, Iris has introduced a new set of summary numbers, called the "Prime" numbers: $+Q_m'$, $-Q_m'$, $+NQN'$, and $-NQN'$. These "Prime" summary numbers are calculated using a similar algorithm to the traditional (normal) numbers with the exception of the polarity evaluation. Thus, the $-Q_m'$ will include all of the pulses that occur during the positive AC half-cycle regardless of the pulse polarity and the $+Q_m'$ will include all of the pulses during the negative AC half-cycle regardless of pulse polarity. Similar adjustments are made for the NQN' values.

You can Toggle between the traditional and Prime (Alternate) summary variables in PDLitePro using the right-click option in the Measurements Window.



F. An...	Freq. (...)	Alt Qm/NQN	C1 NQN+	C1 NQN-	C1 Qm+	C1 Qm-	C2 NQN+	C2 NQN-	C2 Qm+	C2 Qm-	Amb Hum
30	60 Hz, (...)	No	-	-	-	-	8.8	11.9	5.0	7.0	70 %
270	60 Hz, (...)	No	1.7	3.1	-	-	69.3	75.7	35.0	41.0	55 %
150	60 Hz, (...)	No	5.0	7.3	-	-	34.7	45.0	18.0	22.0	55 %
30	60 Hz, (...)	No	-	-	-	-	60.9	70.3	27.0	32.0	55 %
270	60 Hz, (...)	No	2.7	-	-	-	32.0	36.7	48.0	40.0	55 %
150	60 Hz, (...)	No	-	-	-	-	41.3	47.4	22.0	24.0	55 %
30	60 Hz, (...)	No	2.9	-	-	-	15.1	20.2	11.0	20.0	55 %
270	60 Hz, (...)	No	-	-	-	-	22.7	28.3	13.0	14.0	55 %
150	60 Hz, (...)	No	-	-	-	-	8.1	11.9	8.0	9.0	55 %
30	60 Hz, (...)	No	-	-	-	-	10.3	12.6	9.0	9.0	55 %
150	60 Hz, (...)	No	-	-	-	-	39.7	46.5	20.0	23.0	55 %
30	60 Hz, (...)	No	11.4	-	-	-	5.1	4.9	7.0	5.0	55 %

5.4.1 PD in the Slots

It is important to remember that the winding semi-conductive coating is in fact transparent (i.e. it has an effective high resistance) at high frequencies, or more correctly, the skin depth is so thin at high frequencies, that the electromagnetic field from the PD penetrates the semi-conductive coating and interacts with the SSC. The semi-conductive coating is only conductive at 50/60 Hz, which prevents the SSC from seeing high voltage. From a high frequency point of view, the ground plane of the SSC is the ground plane for traveling waves, but from the point of view of 50/60 Hz signals, the semi-conductive coating is the ground plane.

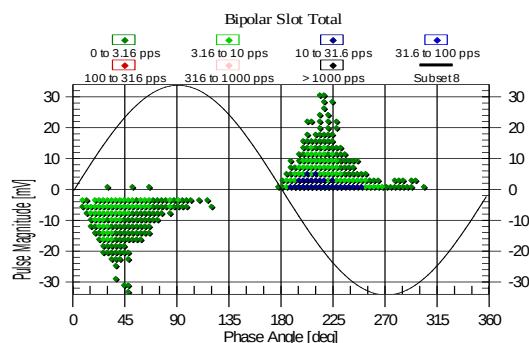
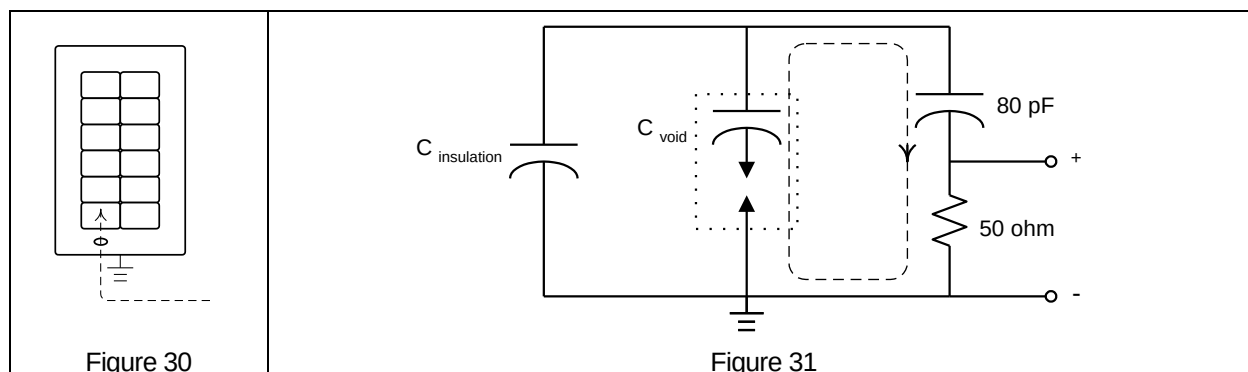


Figure 29: Phase-to-ground PD Activity

In most of the cases, the PD pulses occur due to the voltage difference between the core (ground) and the conductor. With phase-to-ground PD, negative pulses occur from 0° - 90° of the AC cycle, centred on 45° (Figure 29). Positive PD occurs between 180° - 270° of the AC cycle.

These relationships are only valid if the PD sensor is in parallel with the stator winding. In this situation, PD in a void within a stator bar or coil creates a current pulse which (say) travels up to the high voltage phase terminal, passes through the sensor (80 pF) and the 50 Ω load within the instrument, and onto ground (Figure 30 and Figure 31). centred on 225°.



(The PD current is assumed to flow up to the high voltage conductor.)

To complete the current loop (current must flow in a closed loop according to Kirchhoff's current law), the current pulse flows through the ground back to the discharge site (where it capacitively couples to the void). If we take the convention that a current into a resistor is positive in polarity, then a positive voltage is measured with respect to ground in Figure 31 with the current flow as shown. Therefore, when the PD current detected is in *parallel* with the PD site, the measured pulse through the sensor is 180° out-of-phase with the original pulse “through” the void, and the PD will be negative during the positive portion of the cycle.

5.4.2 Prime (Alternate) Numbers

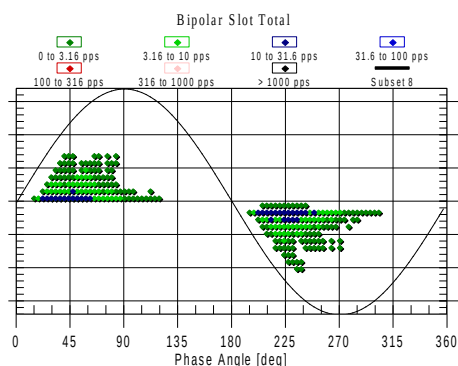


Figure 32: Reversed Phase-to-ground PD

In addition to “typical” PD patterns, when PD is detected with an SSC, it is often observed that if the PD is under the SSC (Figure 32), that the reverse polarity of typical PD pattern is measured.

The PD in parallel with the capacitance of the coil is the capacitance of the transformer winding or other stray capacitances to ground (

Figure 33). In this case, a PD in the coil creates a current loop that flows to the high voltage terminal, through the stray capacitance to ground, through the ground and up through the PD sensors (50 Ω resistor) back to the PD site (

Figure 33).

Since the current flow is now up into the 50 Ω resistor from the groundside, it is in phase with the current through the void. Thus when the PD current detected is in *series* with the PD site, a positive current will be detected during the positive portion of the AC cycle (

Figure 33). The polarity is reversed from the classic pattern. We believe this occurs because if the PD is under the SSC sense line, then current flow will be from the sense line to the 50 Ω resistor in the TGA-S instrument, and back to ground. Thus, instead of the PD being detected in parallel, PD under the SSC sense line is in series and explains why PD may have a reversed polarity in SSC plots (Figure 32).

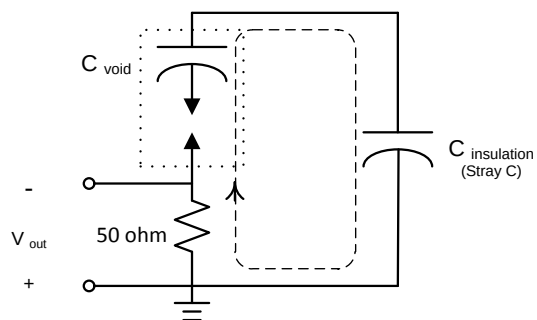


Figure 33



6 REFERENCES

1. IEEE 1434-2001, IEEE Guide to the Measurement of Partial Discharges in Rotating Machinery.
2. G.C. Stone et al, "Electrical Insulation for Rotating Machines", IEEE Press –Wiley, January 2004.
3. Sedding, Howard, "Basics of Rotating Machine Partial Discharge Testing," Canadian Electrical Association - Fourth International Conference on Generator and Motor Partial Discharge Testing Proceedings, Houston, Texas, U.S.A., May 22-24, 1996.
4. Kurtz and G.C. Stone, "Diagnostic testing of generator insulation, Part II - An improved partial discharge test", Canadian Electrical Association Research Report, RP76-17, 1978.
5. Iris Power, "Installation Guide for Epoxy-mica Couplers on Hydrogenerators", 1997.
6. J.F. Lyles, T.E. Goodeve, G.C. Stone, "Using Diagnostic Technology for Identifying Generator Winding Measurement Needs", Hydro Review, June 1993, p58.
7. S.R. Campbell et al, "Practical On-Line PD Tests for Turbine Generators and Motors", IEEE Trans EC, June 1994, p281.
8. G.C. Stone, "Partial Discharge Part XXV: Why PD Calibration is Difficult in Motors and Generators", IEEE Electrical Insulation Magazine, January/February 1998, pp 9-12.
9. V. Warren, "How Much PD is too Much PD", Iris Rotating Machine Conference, March 1998.
10. G.C. Stone et al., "Unusual PD Pulse Phase Distributions in Operating Rotating Machines", IEEE Trans DEI, November 14, 1995, p567
11. V. Warren, "Partial Discharge Testing: A Progress Report", Iris Rotating Machine Conference, June 2004.
12. IEC/TS 60034-27-2 for 1CD, "Part 27-2: On-line partial discharge measurements on the stator winding insulation of rotating electrical machines", International Electrotechnical Commission, 2011.
13. G.C. Stone et al, "Impact of Slot Discharges and Vibration on Stator Winding Life in Large Generators", IEEE Electrical Insulation Magazine September 2008, pp 14-21

7 GLOSSARY

Term	Definition
Ageing	The development of failure mechanisms
Asphalt	Thermoplastic binder to consolidate insulating tape layers
Amplification	An increase in magnitude
Attenuation	A reduction in magnitude
Bar (half coil)	Coils made in half-sections for easier installation in long core or small diameter bore machines - one turn per coil
Bottom bar	The bar further from the stator core bore
Breakdown voltage	The voltage at which a pulse spark will occur across a void
Capacitance	Is the electrical property that stores an electrical charge by impeding a change in voltage
Capacitive couplers	PD sensors that block the 50/60Hz signal and pass the high-frequency PD signals (80pf capacitors)
Capacitive coupling	Travel of a high-frequency signal due to the capacitance property of materials
Phase-to-ground PD pulses	Pulses that occur within the slot portion of the coil and are phase-to-ground voltage dependent - negative clumps near 45° and positive near 225° on the 3-D plot
Contamination	The presence of any conductive or partially conductive material in the endwinding area
Corona	Visible surface partial discharges that occur when the gas adjacent to an exposed conductor ionizes
Cross-coupled PD activity	PD that occurs on another phase from the one that detects it
De-bond	The loss of mechanical strength of a resin which bonds layers of insulting tape together; also called delamination
Delay time	In a directional (BUS) install, the time in nanoseconds it takes for a high frequency pulse to travel from one coupler to the other
Differential (PDA)	A style of sensor installation that uses simultaneous pulse arrival for disturbance separation
Direct (Positive) effect	When an increase in the operating or ambient condition causes an increase in PD – can be attributed to load, temperature, hydrogen pressure or ambient humidity
Directional (BUS)	A style of sensor installation that uses arrival time for disturbance separation
Dispersion	An increase in rise-time due to high frequency pulse traveling waveforms
Electrical breakdown	The theoretical V/mm stress of a gas medium that if exceeded will result in an arc - for air it is $E_{BD} = 3\text{kV (peak)/mm}$ at STP
Electrical slot discharge	Surface PD that occurs within the slot portion of the coil due to inadequate semi-conductive coating
Electrical tracking	The current that flows when a conductive path of surface contamination occurs between two coils at different potentials - frequently across coil blocking



Endwinding (end arm)	That portion of the coil or bar that extends beyond the stator core
Endwinding discharges	Discharges that occur in the end arm area typically between high voltage coils or bars of different phases
Endwinding vibration	100/120Hz or rotational speed frequency vibration of the endwinding due to mechanical stresses
Epoxy	Organic resin used to bond tape layers - usually thermoset
Excitation pulses	Pulses that look like PD, but are really from the thyristor firing of the excitation system – typically low magnitudes and occur at uniform phase locations across the cycle, usually six.
Form-wound coils	Coils designed such that the turn voltage stress is constant at power system frequency which is normally 50 or 60 Hz
Frequency - pulse	The inverse of four times the rise-time ($f = 1 / 4 * \text{rise-time}$)
Girth cracking	A crack around the circumference of the coil at slot exit - normally caused by Thermal Cycling of asphaltic insulation systems
Glass	Fibreglass materials used in insulation systems - these may be glass fibre (roving), glass tape or glass felt materials
Global VPI	When the VPI process is applied to an entire stator after the green coils or bars have been secured inside the stator core
Glow discharge	A type of partial discharge phenomenon characterized by a diffused glow [1]
Grading coating (stress control coating)	A silicon-carbide tape or paint that is applied to the outside portion of the coil/bar starting beyond the slot exit and extending into the endwinding - used for voltage stress control – extent of surface covered is voltage dependent
Green coil	A dry-tape coil which has not yet been subjected to a VPI process or a resin rich coil which has not been pressed and cured
Groundwall insulation	The layers of insulation that are wrapped around the entire coil side or bar to withstand the voltage to ground electric stress
Guard system	Continuous monitoring systems for monitoring PD activity in generators and motors – HydroGuard, BusGuard and TurboGuard
GuardII system	Second generation continuous monitoring systems for monitoring PD activity in generators and motors, expandable for other monitoring technologies such as rotor flux and endwinding vibration
Humidity effect	When there is a substantial change in PD with a change in humidity -- this is most often an indication of surface contamination
Hydrogen pressure effect	When there is a substantial change in PD with a change in hydrogen pressure -- a normal phenomenon, but more apparent with surface PD activity
Impedance	The electrical property that opposes the flow of current
Impedance mismatch	The natural tendency of high-frequency pulses to attenuate or amplify when they traverse from a material of one characteristic impedance level to another
Impregnation	The manufacturing process that forces resin to seep and cure between the layers of tape

Improper impregnation	When a coil or bar is not properly impregnated with organic resin during manufacturing and has voids within the insulation structure
Internal delamination	The de-bonding of tape layers due to thermal stresses
Inductance	The electrical property that produces an electromagnetic field by impeding a change of the current
Insulation failure	A complete electrical discharge that occurs between two metallic electrodes
Inverse (Negative) effect	When an increase in the operating or ambient condition causes a decrease in PD – can be attributed to load, temperature, hydrogen pressure or ambient humidity
Ladder-effect	Result of coil vibration where the portion of the coil adjacent to the core abrades and that near the ventilation ducts does not - the pattern of the damage outside the coil resembles a ladder
Line-end coil	The coil connected to the circuit ring or phase bus - will have maximum phase-to-ground voltage
Load cycling	Frequent changes in the machine's load current due to either real and reactive power
Load effect	When there is a substantial change in PD with a change in load -- if positive predominant as well, this is an indication of loose coils
Loose coils	Coils that are inadequately secured within the slot and thus vibrate due to the mechanical and magnetic stresses in a machine
Mica	The principal insulating dielectric material of stator windings - it is a durable but brittle mineral product
Multi-turn coil	Coils made with several conductor turns
Negative predominance	When the amount of negative phase-to-ground PD, near 45°, is higher than the amount of positive phase-to-ground PD near 225° -- indicative of slot insulation voids near the conductors
No load test	Test at no load or reduced load for determining load effect by comparing results to a full load test
No polarity predominance	When the amount of negative classic PD, near 45°, is approximately equal to the amount of positive classic PD, near 225° -- indicative of voids within the slot insulation bulk
Noise	Pulses originating from outside of the insulation system, or pulses that do not exhibit PD characteristics
Disturbance separation	The technique used to ensure that only machine PD is detected and not PD-like disturbances from sources outside the machine stator winding
Disturbance-like PD	Non-classic PD detected by the machine coupler, which consists of small magnitude pulses occurring across the AC cycle. This is usually not indicative of a problem, and as the winding ages "true" PD will eventually be observed beyond the disturbance-like PD.
NQN	Normalized Quantity Number - represents the area beneath the curve of the 2-D plots weighted by the magnitude of the pulses

Organic bonding material	Made of organic substance used to bond insulating tape layers - asphalt, polyester, and epoxy
Over voltage	When the voltage stress across a void exceeds the breakdown voltage
Ozone	O ₃ - a by-product of electrical discharges in air, aggressive oxidizer
Partial discharge	An incomplete, or partial, electrical discharge that occurs between insulation and either insulation or a metallic electrode
PDA-IV	Partial Discharge Analyzer - test instrument for monitoring PD activity in hydro generators
Phase-to-Phase Discharge	PD that is phase-to-phase voltage dependent and originates outside of the slot from contamination, electrical tracking, or inadequate spacing – pulses clump 30° phase-shifted from the phase-to-ground PD positions (or 15°, 75°, 195°, and 255°) with corresponding opposite polarity pulse patterns in the two phases between which PD is occurring
Polarity predominance	When the Q _m of one polarity is greater than that of the other polarity, or there is a separation of the two lines on the 2-D plot
Polyester	Organic resin used to bond tape layers - usually a thermoset material
Positive predominance	When the amount of positive phase-to-ground PD, around 225°, is higher than the amount of negative phase-to-ground PD, near 45° -- usually indicative of surface activity
Pseudo-glow discharge	A type of partial discharge characterized by current pulses of relatively small amplitude, and generally, a long rise time
Q _m	Peak PD magnitude in mV measured at a repetition rate of 10 pulses per second
Radiation	Energy emitted from a central point in the form of waves or particles
Resin-loaded tapes	Tapes that are epoxy-loaded and are directly applied to the bars - they are cured with heat and pressure
Rise-time	The time required for a pulse to increase from 10% to 90% of its peak magnitude
RTD	Resistive temperature detector - used for monitoring stator winding temperatures
Semi-conducting slot coating	A carbon-loaded tape or paint that is applied to the outside of the portion of the coil or bar that will be in contact with the core - used for voltage stress control for voltage ratings of 6 kV and above and for motors with lower voltage ratings if fed from a variable voltage and frequency converter
Side packing	The material used to fill up the space between a coil or bar side and the stator core - can be rigid or conformable, and is usually semi-conductive for voltage ratings of 6 kV and above and for motors with lower voltage ratings if fed from a variable voltage and frequency converter
Signal-to-noise ratio (SNR)	The logarithmic ratio of the PD signal to noise. A high SNR is necessary for noise separation.
Slot discharges	Discharges that occur on a random basis involving a large surface of coil or bar, separated from the grounded core by an air gap – positive predominance
Slot portion	The portion of the coil/bar located within the stator core section



Stator slot couplers (SSC)	Wideband antenna installed beneath the wedges or between the bars, which is capable of distinguishing PD based on pulse characteristics
Temperature effect	When a change in temperature affects the PD levels
TGA	Turbine Generator Analyzer - test instrument for monitoring PD activity in turbine generators and motors
Thermoplastic	A coil insulating system the binding materials of which become somewhat soft and pliable with an increase in temperature – asphaltic-mica is a classic example
Thermoset	A coil insulating system the binding materials of which set up rigidly and are relatively unaffected by normal operating temperatures – epoxy-mica is an example
Time-of-arrival	A disturbance separation technique that compares the arrival of pulses at two different couplers to determine the direction of the PD source
Top bar	The bar closest to the armature
Trac monitor	Continuous monitoring systems for monitoring PD activity in generators and motors – HydroTrac, BusTrac and PDTrac
BusTracII, HydroTracII and PDTracII monitor	Second generation continuous monitoring systems for monitoring PD activity in generators and motors
Transmission	Travel of an electrical current through a conductor
Turn insulation	A dedicated layer of insulation surrounding each turn in a multi-turn coil
Vacuum pressure impregnation (VPI)	An impregnation process that involves creating a vacuum, injecting resin, and applying pressure - can be done on a batch of coils or the entire stator
Voltage stress control	The application of semi-conductive and grading materials to control electrical stresses on the surface of coil insulation